

Bachelors with Mathematics as Major 5th Semester MMT522J3: Mathematics/Applied Mathematics: FOURIER AND LAPLACE TRANSFORM	
Credits: 4 THEORY + 2 TUTORIAL	Theory: 60 Hours & Tutorial: 30 Hours
Course Objectives: To develop skill in students about i) Fourier series, Fourier and Laplace transforms as a tool to solve various problems of Mathematics. ii) Integral transforms to be used in the field of applied Mathematics and especially in the field of physics and electronics to express periodic functions that comprise communication signal in waveform. Course Outcome: After the completion of this course, students shall be able to use Fourier and Laplace transforms to solve the differential equations and to understand signal processing in frequency and time domain.	
Theory: 4 Credi	
Unit- I Fourier Series: Introduction, Periodic functions: Properties, Even & Odd functions. Special waveforms: Square wave, Sawtooth wave, and Triangular wave. Euler's Formulae for Fourier Series, Fourier Series for functions of period 2π, Fourier Series for functions of period $2L$, Dirichlet's conditions, Sum of Fourier series. If $f(x)$ is bounded and integrable function on $(-\pi, \pi)$ and if a_n, b_n are its Fourier coefficients, then $\sum a_n^2 + b_n^2$ converges. Half Range Series for sine and cosine functions, examples. Riemann Lebesque theorem.	
Unit – II The Fourier Transform. Periodic functions, Definition and examples of Fourier series, Dirichlet's conditions, determination of Fourier coefficients, even and odd functions and their Fourier expansion, change of interval, half range series. Fourier transform, inverse Fourier transform, Fourier sine and cosine transforms and their inversion, properties of Fourier transforms. Fourier transform of the derivative, convolution theorem, discrete Fourier transform and fast Fourier transform and their properties.	
Unit – III Definition, Laplace transform of elementary functions, Properties of Laplace transforms viz Linearity, translation, Change of Scale property etc. Laplace transform as periodic functions, Dirac-Delta function, Inverse Laplace transform. Laplace transform for derivatives, Laplace transform for integrals, Convolution theorem. Solution of ordinary differential equations with constant coefficients. Applications of partial differential equations.	
Unit – IV Application of Laplace transform to differential equation and integral equation. Application of Laplace transform to boundary value problems. Electrical circuits, dynamics, Beams, Heat conduction equations and wave equations.	
Tutorials: 2 Credits	
Unit – V Problems based on unit-I and unit-II.	
Unit – VI Problems based on unit-III and unit-IV.	
Recommended Books: 1. Ruel V.Churchill, Fourier Series & Boundary Value Problems, 8th Edition McGraw Hill Education 2011. 2. Davies, Brian, Integral Transforms and Their Applications, Springer, 2002. 3. Erwin, Kreysgiz, Advanced Engineering Mathematics, John Willey & Sons. 10th Edition, 2011. 4. K.S. Rao, Introduction to Partial Differential Equations, K.S. Rao, PHI, India. 5. Murrey R. Spiegel, Laplace Transforms, Schaum's outline series. 6. I. N. Sneddon: The use of Integral Transforms, McGraw-Hill, Singapore 1972. 7. R. R. Goldberg, Fourier Transforms, Cambridge University Press, 1961. 8. D. Brain, Integral Transforms and their applications, Springer, 2002.	



LAPACE AND FOURIER TRANSFORM (MMT522J3)



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Chapter 1

Fourier Series and Transforms

1.1 Introduction to Fourier Series

1.1.1 Periodic Functions

The central objects of study for Fourier series are periodic functions.

Definition 1.1.1 (Periodic Function). A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be **periodic** with a **period** $T > 0$ if for every x in its domain,

$$f(x + T) = f(x)$$

The smallest positive value of T for which this relation holds is called the **fundamental period** of the function.

Example 1.1.2. 1. The function $f(x) = \sin(x)$ is periodic with periods $2\pi, 4\pi, 6\pi, \dots$. Its fundamental period is $T = 2\pi$, since this is the smallest positive value for which $\sin(x + T) = \sin(x)$.

2. Similarly, $g(x) = \cos(x)$ has a fundamental period of $T = 2\pi$.

3. The function $h(x) = \tan(x)$ has a fundamental period of $T = \pi$.

4. A constant function, $f(x) = c$, is periodic for any period $T > 0$. It does not have a fundamental period.

Properties of Periodic Functions

Let $f(x)$ and $g(x)$ be periodic functions with a common period T .

1. **Linear Combination:** The sum $h(x) = c_1 f(x) + c_2 g(x)$, where c_1 and c_2 are constants, is also periodic with period T .
2. **Product and Quotient:** The product $h(x) = f(x)g(x)$ and the quotient $q(x) = f(x)/g(x)$ (where $g(x) \neq 0$) are also periodic with period T .
3. **Horizontal Scaling:** If $f(x)$ has a period T , then the function $g(x) = f(kx)$ for some constant $k \neq 0$ is periodic with period $T/|k|$. For instance, since $\sin(x)$ has period 2π , the function $\sin(3x)$ has period $2\pi/3$.
4. **Least Common Multiple Period:** If $f(x)$ has period T_1 and $g(x)$ has period T_2 , and the ratio T_1/T_2 is a rational number, then the sum $f(x) + g(x)$ is periodic. Its fundamental period is the least common multiple (LCM) of T_1 and T_2 .

1.1.2 Even and Odd Functions

Recognizing the symmetry of a function can greatly simplify the calculation of its Fourier series.

Definition 1.1.3 (Even Function). A function $f(x)$ is **even** if for every x in its domain,

$$f(-x) = f(x)$$

Geometrically, the graph of an even function is symmetric with respect to the y-axis.

Examples: $x^2, x^4, \cos(x), |x|$.

Definition 1.1.4 (Odd Function). A function $f(x)$ is **odd** if for every x in its domain,

$$f(-x) = -f(x)$$

Geometrically, the graph of an odd function is symmetric with respect to the origin.

Examples: $x, x^3, \sin(x), \tan(x)$.

Properties of Even and Odd Functions

These properties are essential for simplifying Fourier coefficient calculations.

1. Algebraic Combinations:

- (Even) $\times$ (Even) = Even
- (Odd) $\times$ (Odd) = Even
- (Even) $\times$ (Odd) = Odd
- (Even) $\pm$ (Even) = Even
- (Odd) $\pm$ (Odd) = Odd

2. **Decomposition:** Any function $f(x)$ can be expressed as the sum of an even function and an odd function:

$$f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{Even Part}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{Odd Part}}$$

3. **Integration over Symmetric Intervals:** Let $[-L, L]$ be an interval symmetric about the origin.

- If $f(x)$ is an **odd** function, then $\int_{-L}^L f(x) dx = 0$.
- If $f(x)$ is an **even** function, then $\int_{-L}^L f(x) dx = 2 \int_0^L f(x) dx$.

1.1.3 Special Periodic Waveforms

In engineering and physics, several fundamental periodic functions serve as building blocks for more complex signals. We define three of the most common ones here. For simplicity, we will define them over an interval of length 2π and state their fundamental period.

Square Wave

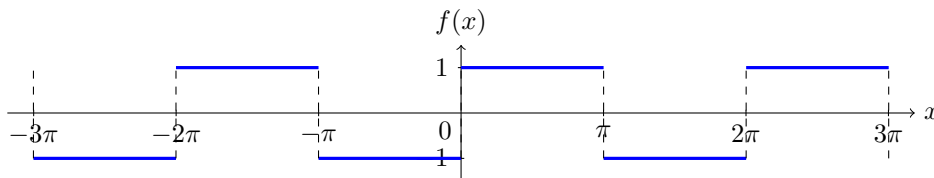
A square wave is a non-sinusoidal periodic waveform that alternates between two fixed voltage or current levels.

Definition 1.1.5. A standard square wave with period $T = 2\pi$ can be defined over the interval $(-\pi, \pi)$ as:

$$f(x) = \begin{cases} -1 & \text{if } -\pi < x < 0 \\ 1 & \text{if } 0 < x < \pi \end{cases}$$

and is extended periodically by the relation $f(x + 2\pi) = f(x)$. The function is typically left undefined at the points of discontinuity ($x = 0, \pm\pi, \pm2\pi, \dots$).

Symmetry: The square wave as defined above is an **odd function**, since $f(-x) = -f(x)$.



Sawtooth Wave

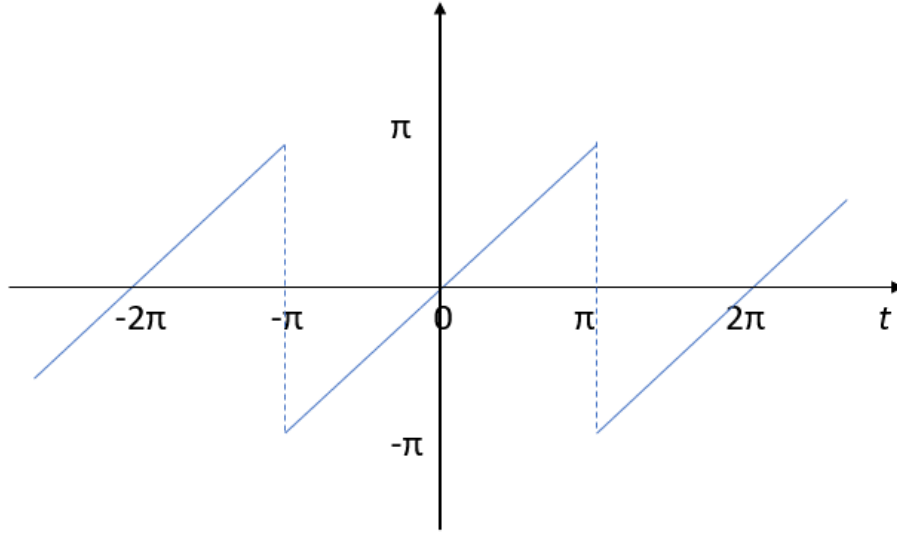
A sawtooth wave is a waveform that ramps up linearly and then sharply falls to its starting value, or vice-versa. It resembles the teeth of a saw.

Definition 1.1.6. A standard sawtooth wave with period $T = 2\pi$ can be defined over the interval $(-\pi, \pi)$ as:

$$f(x) = x$$

and is extended periodically by the relation $f(x + 2\pi) = f(x)$.

Symmetry: The sawtooth wave as defined above is an **odd function**.



Triangular Wave

A triangular wave is another non-sinusoidal waveform characterized by a linear rise and fall, creating a triangular shape.

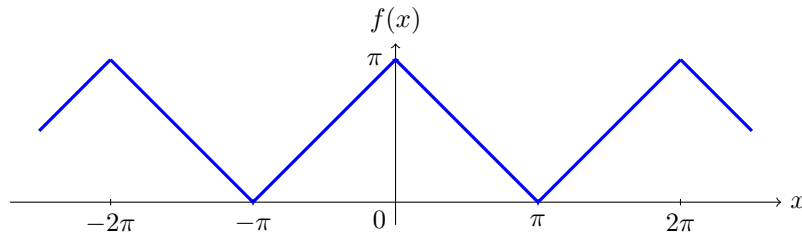
Definition 1.1.7. A standard triangular wave with period $T = 2\pi$ can be defined on the interval $[-\pi, \pi]$ using the absolute value function:

$$f(x) = \pi - |x|$$

and is extended periodically by the relation $f(x + 2\pi) = f(x)$. This can also be written piecewise as:

$$f(x) = \begin{cases} \pi + x & \text{if } -\pi \leq x < 0 \\ \pi - x & \text{if } 0 \leq x \leq \pi \end{cases}$$

Symmetry: The triangular wave as defined above is an **even function**, since $f(-x) = \pi - |-x| = \pi - |x| = f(x)$.



1.2 The Fourier Series Representation

The central goal of Fourier analysis is to represent a periodic function $f(x)$ as an infinite series of sine and cosine functions. This trigonometric series is known as the Fourier series.

1.2.1 Fourier Series for Functions of Period $2L$

Let us consider a function $f(x)$ that is periodic with a general period $T = 2L$. The Fourier series representation of $f(x)$ on an interval of length $2L$, for example $[-L, L]$, is given by:

Definition 1.2.1 (Fourier Series). The Fourier series of a function $f(x)$ defined on the interval $[-L, L]$ is given by

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right)$$

where the coefficients a_0 , a_n , and b_n are the **Fourier coefficients**. The symbol $\sim$ is used to indicate that the series on the right corresponds to the function $f(x)$, without making any claims about convergence at this stage.

Remark 1.2.2. The term $\frac{a_0}{2}$ is the average value (or DC component) of the function over one period. The reason for the factor of $1/2$ is for notational convenience, as it allows the general formula for a_n to also work for $n = 0$.

1.2.2 Euler's Formulae for the Fourier Coefficients

The Fourier coefficients are determined by a set of integral formulas known as Euler's formulae. These are derived by exploiting the **orthogonality properties** of sine and cosine functions over the interval $[-L, L]$. The key integral relations are:

$$\begin{aligned} \bullet \int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx &= \begin{cases} 0 & \text{if } m \neq n \\ L & \text{if } m = n \neq 0 \\ 2L & \text{if } m = n = 0 \end{cases} \\ \bullet \int_{-L}^L \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx &= \begin{cases} 0 & \text{if } m \neq n \\ L & \text{if } m = n \neq 0 \end{cases} \\ \bullet \int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx &= 0 \text{ for all integers } m, n. \end{aligned}$$

By integrating the Fourier series term-by-term (a process that can be rigorously justified), we arrive at Euler's formulae.

Calculating the Coefficients

- **To find a_0 :** Integrate the Fourier series from $-L$ to L . Due to orthogonality, every sine and cosine term integrates to zero.

$$\int_{-L}^L f(x) dx = \int_{-L}^L \frac{a_0}{2} dx = \frac{a_0}{2} (2L) = a_0 L$$

- **To find a_n ($n \geq 1$):** Multiply the entire series by $\cos\left(\frac{n\pi x}{L}\right)$ and integrate from $-L$ to L . All terms integrate to zero except for the one where $n = m$.

$$\int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx = a_n \int_{-L}^L \cos^2\left(\frac{n\pi x}{L}\right) dx = a_n L$$

- **To find b_n ($n \geq 1$):** Multiply the series by $\sin\left(\frac{n\pi x}{L}\right)$ and integrate. All terms integrate to zero except for the one where $n = m$.

$$\int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx = b_n \int_{-L}^L \sin^2\left(\frac{n\pi x}{L}\right) dx = b_n L$$

This leads to the definitive formulas:

Theorem 1.2.3 (Euler's Formulae). *For a function $f(x)$ with period $2L$, the Fourier coefficients are given by:*

$$\begin{aligned} a_0 &= \frac{1}{L} \int_{-L}^L f(x) dx \\ a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \quad (n = 1, 2, 3, \dots) \\ b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \quad (n = 1, 2, 3, \dots) \end{aligned}$$

1.2.3 Fourier Series for Functions of Period 2π

A very common and important case is when the period of the function is $T = 2\pi$. This corresponds to setting $L = \pi$ in the general formulas. The term $\frac{n\pi x}{L}$ simplifies to $\frac{n\pi x}{\pi} = nx$.

The Fourier series for a function $f(x)$ with period 2π on the interval $[-\pi, \pi]$ is:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

The corresponding Euler's formulae for the coefficients become:

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \\ a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx \quad (n = 1, 2, 3, \dots) \\ b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx \quad (n = 1, 2, 3, \dots) \end{aligned}$$

Remark 1.2.4 (Interval of Integration). The interval of integration must have a length equal to the period $T = 2L$. While $[-L, L]$ is common, any interval such as $[0, 2L]$ or $[c, c + 2L]$ can be used. For a period of 2π , the interval $[0, 2\pi]$ is frequently used instead of $[-\pi, \pi]$. The choice often depends on the definition of the function.

Example 1.2.5 (Fourier Series of a Square Wave). Find the Fourier series for the square wave function $f(x)$ with period $T = 2\pi$ defined on $(-\pi, \pi)$ as:

$$f(x) = \begin{cases} -1 & \text{if } -\pi < x < 0 \\ 1 & \text{if } 0 < x < \pi \end{cases}$$

Solution. The function is periodic with period 2π , so we set $L = \pi$. By inspection, $f(-x) = -f(x)$, so the function is odd. Consequently, all cosine coefficients are zero: $a_n = 0$ for $n \geq 0$. We need only compute the sine coefficients, b_n . Since $f(x) \sin(nx)$ is an even function (a product of two odd functions), we can write:

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx = \frac{2}{\pi} \int_0^{\pi} (1) \sin(nx) dx \\ &= \frac{2}{\pi} \left[-\frac{\cos(nx)}{n} \right]_0^{\pi} = \frac{2}{n\pi} (1 - \cos(n\pi)) \end{aligned}$$

The term $(1 - \cos(n\pi))$ is 0 if n is even, and 2 if n is odd. Thus, $b_n = 0$ for even n and $b_n = \frac{4}{n\pi}$ for odd n . The resulting Fourier series is composed only of sine terms with odd harmonics:

$$f(x) \sim \sum_{n=1,3,5,\dots}^{\infty} \frac{4}{n\pi} \sin(nx) = \frac{4}{\pi} \left(\sin(x) + \frac{1}{3} \sin(3x) + \frac{1}{5} \sin(5x) + \dots \right)$$

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Example 1.2.6 (Fourier Series of a Triangular Wave). Find the Fourier series for the triangular wave function $f(x) = \pi - |x|$ with period $T = 2\pi$ on the interval $[-\pi, \pi]$.

Solution. The function has period 2π ($L = \pi$) and is an even function since $f(-x) = \pi - |-x| = \pi - |x| = f(x)$. Therefore, all sine coefficients are zero: $b_n = 0$ for all $n \geq 1$. The DC component, a_0 , is calculated as:

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} (\pi - |x|) dx = \frac{2}{\pi} \int_0^{\pi} (\pi - x) dx = \frac{2}{\pi} \left[\pi x - \frac{x^2}{2} \right]_0^{\pi} = \pi$$

For the cosine coefficients a_n ($n \geq 1$), the integrand $(\pi - |x|) \cos(nx)$ is even:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} (\pi - |x|) \cos(nx) dx = \frac{2}{\pi} \int_0^{\pi} (\pi - x) \cos(nx) dx$$

Using integration by parts, we find:

$$\begin{aligned} a_n &= \frac{2}{\pi} \left(\left[(\pi - x) \frac{\sin(nx)}{n} \right]_0^{\pi} + \frac{1}{n} \int_0^{\pi} \sin(nx) dx \right) \\ &= \frac{2}{n\pi} \left[-\frac{\cos(nx)}{n} \right]_0^{\pi} = \frac{2}{n^2\pi} (1 - \cos(n\pi)) = \frac{2}{n^2\pi} (1 - (-1)^n) \end{aligned}$$

This yields $a_n = 0$ for even n and $a_n = \frac{4}{n^2\pi}$ for odd n . The Fourier series is:

$$\begin{aligned} f(x) &\sim \frac{a_0}{2} + \sum_{n=1,3,5,\dots}^{\infty} a_n \cos(nx) = \frac{\pi}{2} + \frac{4}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{\cos(nx)}{n^2} \\ f(x) &\sim \frac{\pi}{2} + \frac{4}{\pi} \left(\cos(x) + \frac{1}{9} \cos(3x) + \frac{1}{25} \cos(5x) + \dots \right) \end{aligned}$$

■

Example 1.2.7 (Fourier Series for Period $2L$). Find the Fourier series for the function $f(x) = x$ defined on the interval $[0, 2)$ with period $T = 2$.

Solution. The function has period $T = 2$, so $2L = 2$ which implies $L = 1$. We compute the coefficients by integrating over the interval $[0, 2]$. The DC component a_0 is:

$$a_0 = \frac{1}{1} \int_0^2 x dx = \left[\frac{x^2}{2} \right]_0^2 = 2$$

The cosine coefficients a_n for $n \geq 1$ are:

$$a_n = \int_0^2 x \cos(n\pi x) dx = \left[\frac{x \sin(n\pi x)}{n\pi} \right]_0^2 - \int_0^2 \frac{\sin(n\pi x)}{n\pi} dx = 0 - \left[\frac{\cos(n\pi x)}{n^2\pi^2} \right]_0^2 = 0$$

The sine coefficients b_n for $n \geq 1$ are:

$$\begin{aligned} b_n &= \int_0^2 x \sin(n\pi x) dx = \left[\frac{-x \cos(n\pi x)}{n\pi} \right]_0^2 + \int_0^2 \frac{\cos(n\pi x)}{n\pi} dx \\ &= -\frac{2 \cos(2n\pi)}{n\pi} + \left[\frac{\sin(n\pi x)}{n^2\pi^2} \right]_0^2 = -\frac{2}{n\pi} \end{aligned}$$

Assembling the series with $a_0 = 2$, $a_n = 0$, and $b_n = -2/(n\pi)$:

$$\begin{aligned} f(x) &\sim \frac{a_0}{2} + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) = \frac{2}{2} + \sum_{n=1}^{\infty} \left(-\frac{2}{n\pi}\right) \sin(n\pi x) \\ f(x) &\sim 1 - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{\sin(n\pi x)}{n} = 1 - \frac{2}{\pi} \left(\sin(\pi x) + \frac{1}{2} \sin(2\pi x) + \frac{1}{3} \sin(3\pi x) + \dots \right) \end{aligned}$$

■

Exercise 1.2.8. Find the Fourier series expansion for each of the following functions:

1. Find the Fourier series of the square wave function defined by:

$$f(x) = \begin{cases} \pi + x & \text{for } -\pi < x < 0 \\ 0 & \text{for } 0 \leq x < \pi, \end{cases} \quad f(x + 2\pi) = f(x).$$

with period 2π .

2. Determine the Fourier series of the sawtooth wave function:

$$f(x) = \begin{cases} 2 + x & \text{for } -2 \leq x \leq 0 \\ 2 - x & \text{for } 0 \leq x \leq 2, \end{cases} \quad f(x + 4) = f(x).$$

with period 2π .

3. Compute the Fourier series of the triangular wave function:

$$f(x) = \begin{cases} \pi + x & \text{for } -\pi \leq x \leq 0 \\ \pi - x & \text{for } 0 \leq x \leq \pi \end{cases}$$

with period 2π .

For each function:

- Calculate the Fourier coefficients a_0 , a_n , and b_n
- Write the complete Fourier series expansion
- State the value to which the series converges at points of discontinuity (if any)

1.3 Fourier Series of Even and Odd Functions

Definition 1.3.1 (Even Function). A function $f(x)$ is **even** if for all x in its domain:

$$f(-x) = f(x)$$

The graph of an even function is symmetric about the y-axis.

Definition 1.3.2 (Odd Function). A function $f(x)$ is **odd** if for all x in its domain:

$$f(-x) = -f(x)$$

The graph of an odd function is symmetric about the origin.

Fourier Series Simplifications

For a function $f(x)$ with period $2L$:

- If $f(x)$ is **even**, then all b_n coefficients vanish:

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx = 0$$

The Fourier series becomes a **Fourier cosine series**:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

where

$$a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

- If $f(x)$ is **odd**, then all a_n coefficients vanish:

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx = 0$$

The Fourier series becomes a **Fourier sine series**:

$$f(x) \sim \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

where

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

Example 1.3.3 (Even Function). Find the Fourier series of $f(x) = x^2$ on $[-\pi, \pi]$.

Solution: Since $f(-x) = (-x)^2 = x^2 = f(x)$, the function is even. We compute the coefficients:

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x^2 dx = \frac{2}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi} = \frac{2\pi^2}{3}$$

For $n \geq 1$:

$$a_n = \frac{2}{\pi} \int_0^{\pi} x^2 \cos(nx) dx$$

Using integration by parts twice:

$$a_n = \frac{2}{\pi} \left[\frac{x^2 \sin(nx)}{n} + \frac{2x \cos(nx)}{n^2} - \frac{2 \sin(nx)}{n^3} \right]_0^{\pi} = \frac{2}{\pi} \left(\frac{2\pi \cos(n\pi)}{n^2} \right) = \frac{4(-1)^n}{n^2}$$

Thus, the Fourier series is:

$$x^2 \sim \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2} \cos(nx)$$

Example 1.3.4 (Odd Function). Find the Fourier series of $f(x) = x$ on $[-\pi, \pi]$.

Solution: Since $f(-x) = -x = -f(x)$, the function is odd. We compute the coefficients:

$$b_n = \frac{2}{\pi} \int_0^{\pi} x \sin(nx) dx$$

Using integration by parts:

$$b_n = \frac{2}{\pi} \left[-\frac{x \cos(nx)}{n} + \frac{\sin(nx)}{n^2} \right]_0^{\pi} = \frac{2}{\pi} \left(-\frac{\pi \cos(n\pi)}{n} \right) = \frac{2(-1)^{n+1}}{n}$$

Thus, the Fourier series is:

$$x \sim \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n} \sin(nx)$$

Example 1.3.5 (Mixed Function). Find the Fourier series of $f(x) = \begin{cases} x & \text{for } 0 \leq x \leq \pi \\ 0 & \text{for } -\pi \leq x < 0 \end{cases}$ on $[-\pi, \pi]$.

Solution: This function is neither even nor odd. We compute all coefficients:

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} x dx = \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} = \frac{\pi}{2}$$

For $n \geq 1$:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx = \frac{1}{\pi} \int_0^{\pi} x \cos(nx) dx$$

Using integration by parts:

$$a_n = \frac{1}{\pi} \left[\frac{x \sin(nx)}{n} + \frac{\cos(nx)}{n^2} \right]_0^{\pi} = \frac{1}{\pi} \left(\frac{\cos(n\pi)}{n^2} - \frac{1}{n^2} \right) = \frac{(-1)^n - 1}{\pi n^2}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx = \frac{1}{\pi} \int_0^{\pi} x \sin(nx) dx$$

Using integration by parts:

$$b_n = \frac{1}{\pi} \left[-\frac{x \cos(nx)}{n} + \frac{\sin(nx)}{n^2} \right]_0^{\pi} = \frac{1}{\pi} \left(-\frac{\pi \cos(n\pi)}{n} \right) = \frac{(-1)^{n+1}}{n}$$

Thus, the Fourier series is:

$$f(x) \sim \frac{\pi}{4} + \sum_{n=1}^{\infty} \left[\frac{(-1)^n - 1}{\pi n^2} \cos(nx) + \frac{(-1)^{n+1}}{n} \sin(nx) \right]$$

1.4 Convergence of Fourier Series

So far, we have used the symbol $\sim$ to relate a function to its Fourier series. This was a deliberate choice because we have not yet established the conditions under which the infinite series actually converges, and if it does, whether it converges to the original function $f(x)$. The following sections address this crucial question.

1.4.1 Dirichlet's Conditions

The conditions that guarantee the convergence of a Fourier series are known as Dirichlet's conditions. Most periodic functions encountered in science and engineering satisfy these conditions.

Definition 1.4.1 (Dirichlet's Conditions). A function $f(x)$ defined on the interval $[-L, L]$ is said to satisfy **Dirichlet's conditions** if it meets the following criteria:

1. $f(x)$ is single-valued and periodic with period $2L$.
2. $f(x)$ has a finite number of finite discontinuities within one period. (The function is **piecewise continuous**).
3. $f(x)$ has a finite number of maxima and minima within one period. (The function is of **bounded variation**, or more simply, **piecewise monotonic**).

Remark 1.4.2. In simpler terms, these conditions mean that the function must be "well-behaved". It cannot have an infinite number of wiggles or jumps in a single period, nor can it shoot off to infinity. The square wave, sawtooth wave, and triangular wave all satisfy Dirichlet's conditions.

Riemann-Lebesgue Lemma

Theorem 1.4.3 (Riemann-Lebesgue Lemma). Let f be an absolutely integrable function on the interval $[a, b]$, i.e.,

$$\int_a^b |f(t)| dt < \infty.$$

Then:

$$\lim_{n \rightarrow \infty} \int_a^b f(t) \sin(nt) dt = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \int_a^b f(t) \cos(nt) dt = 0.$$

In particular, for Fourier coefficients of an integrable periodic function, we have:

$$\lim_{n \rightarrow \infty} a_n = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} b_n = 0.$$

Proof. We prove the result for the sine integral; the cosine case is analogous.

First consider a step function $\phi(t)$ on $[a, b]$, which is constant on subintervals:

$$\phi(t) = c_j \quad \text{for } t \in (t_{j-1}, t_j), \quad j = 1, \dots, m.$$

Then:

$$\int_a^b \phi(t) \sin(nt) dt = \sum_{j=1}^m c_j \int_{t_{j-1}}^{t_j} \sin(nt) dt = \sum_{j=1}^m \frac{c_j}{n} [-\cos(nt)]_{t_{j-1}}^{t_j}.$$

Since $|\cos(nt)| \leq 1$, we have:

$$\left| \int_a^b \phi(t) \sin(nt) dt \right| \leq \sum_{j=1}^m \frac{2|c_j|}{n} \leq \frac{2M}{n} \sum_{j=1}^m (t_j - t_{j-1}) = \frac{2M(b-a)}{n},$$

where $M = \max_j |c_j|$. Clearly, this tends to 0 as $n \rightarrow \infty$.

For any integrable function f and any $\epsilon > 0$, there exists a step function ϕ such that:

$$\int_a^b |f(t) - \phi(t)| dt < \frac{\epsilon}{2}.$$

Then:

$$\left| \int_a^b f(t) \sin(nt) dt \right| \leq \left| \int_a^b [f(t) - \phi(t)] \sin(nt) dt \right| + \left| \int_a^b \phi(t) \sin(nt) dt \right|.$$

The first term is bounded by:

$$\int_a^b |f(t) - \phi(t)| dt < \frac{\epsilon}{2}.$$

For the second term, since ϕ is a step function, by Step 1 there exists n_0 such that for all $n > n_0$:

$$\left| \int_a^b \phi(t) \sin(nt) dt \right| < \frac{\epsilon}{2}.$$

Hence, for $n > n_0$:

$$\left| \int_a^b f(t) \sin(nt) dt \right| < \epsilon,$$

which proves the result.

The application to Fourier coefficients follows by taking:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt) dt, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt,$$

and applying the lemma with $n = n$. ■

1.4.2 The Sum of the Fourier Series (Convergence Theorem)

If a function satisfies Dirichlet's conditions, its Fourier series is guaranteed to converge. The value to which it converges is given by the following theorem.

Theorem 1.4.4 (Fourier Convergence Theorem). *Let $f(x)$ be a periodic function with period $2L$ that satisfies Dirichlet's conditions. The Fourier series of $f(x)$ converges:*

1. to $f(x)$ at all points where $f(x)$ is continuous.
2. to the average of the left-hand and right-hand limits at any point of discontinuity. That is, at a point of discontinuity x_0 , the series converges to:

$$\frac{f(x_0^+) + f(x_0^-)}{2}$$

where $f(x_0^+) = \lim_{h \rightarrow 0^+} f(x_0 + h)$ and $f(x_0^-) = \lim_{h \rightarrow 0^+} f(x_0 - h)$.

Proof. We present the proof for a function with period 2π ; the case for period $2L$ follows by a change of variable.

Let $S_N(x)$ be the N -th partial sum of the Fourier series of f :

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^N (a_n \cos(nx) + b_n \sin(nx)).$$

Substituting the expressions for the Fourier coefficients:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt) dt, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt,$$

we obtain:

$$S_N(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \left[\frac{1}{2} + \sum_{n=1}^N \cos(n(t-x)) \right] dt.$$

The expression in brackets is the Dirichlet kernel:

$$D_N(u) = \frac{1}{2} + \sum_{n=1}^N \cos(nu) = \frac{\sin((N + \frac{1}{2})u)}{2 \sin(u/2)}.$$

Thus,

$$S_N(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) D_N(t-x) dt.$$

Making the substitution $u = t - x$, and using the periodicity of the functions, we get:

$$S_N(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x+u) D_N(u) du = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x+u) \frac{\sin((N + \frac{1}{2})u)}{\sin(u/2)} du.$$

Since the integrand is even in u , we may write:

$$S_N(x) = \frac{1}{2\pi} \int_0^{\pi} [f(x+u) + f(x-u)] \frac{\sin((N + \frac{1}{2})u)}{\sin(u/2)} du.$$

Noting that $\int_0^{\pi} \frac{\sin kx}{x} = \pi$, we consider the difference:

$$S_N(x) - \frac{f(x^+) + f(x^-)}{2} = \frac{1}{2\pi} \int_0^{\pi} \phi(u) \frac{\sin((N + \frac{1}{2})u)}{\sin(u/2)} du,$$

where $\phi(u) = f(x+u) + f(x-u) - (f(x^+) + f(x^-))$.

Since f satisfies Dirichlet's conditions, $\phi(u)/\sin(u/2)$ is absolutely integrable on $[0, \pi]$. By the Riemann–Lebesgue Lemma:

$$\lim_{N \rightarrow \infty} \int_0^{\pi} \phi(u) \frac{\sin((N + \frac{1}{2})u)}{\sin(u/2)} du = 0.$$

Therefore,

$$\lim_{N \rightarrow \infty} S_N(x) = \frac{f(x^+) + f(x^-)}{2}.$$

If f is continuous at x , then $f(x^+) = f(x^-) = f(x)$, and the series converges to $f(x)$. ■

Example 1.4.5 (Convergence of the Square Wave). Consider the square wave on $(-\pi, \pi)$ from the previous examples.

$$f(x) = \begin{cases} -1 & \text{if } -\pi < x < 0 \\ 1 & \text{if } 0 < x < \pi \end{cases}$$

The function is continuous everywhere except at $x = k\pi$ for integers k .

- At a point of continuity, like $x = \pi/2$, the Fourier series converges to $f(\pi/2) = 1$.
- At the discontinuity $x = 0$, the left-hand limit is $f(0^-) = -1$ and the right-hand limit is $f(0^+) = 1$. The series converges to the average:

$$\frac{f(0^+) + f(0^-)}{2} = \frac{1 + (-1)}{2} = 0$$

This is a remarkable result. The series, which consists of only sine terms (all of which are zero at $x = 0$), correctly converges to the midpoint of the jump.

Example 1.4.6. Consider the function $f(x) = x$ for $-\pi < x < \pi$.

1. Find the Fourier series for $f(x)$.
2. Discuss the convergence of the Fourier series at $x = \pi/2$, $x = \pi$, and $x = -\pi$.

Solution. 1. The function $f(x) = x$ is an odd function. Therefore, $a_n = 0$ for all n . We only need to calculate b_n . The period of the function is $2L = 2\pi$, so $L = \pi$.

$$\begin{aligned} b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(nx) dx \end{aligned}$$

We integrate by parts with $u = x$, $dv = \sin(nx)dx$, so $du = dx$, $v = -\frac{1}{n} \cos(nx)$.

$$\begin{aligned} b_n &= \frac{1}{\pi} \left[-\frac{x}{n} \cos(nx) \right]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} -\frac{1}{n} \cos(nx) dx \\ &= \frac{1}{\pi} \left[\left(-\frac{\pi}{n} \cos(n\pi) \right) - \left(\frac{\pi}{n} \cos(-n\pi) \right) \right] + \frac{1}{n\pi} \int_{-\pi}^{\pi} \cos(nx) dx \\ &= \frac{1}{\pi} \left[-\frac{\pi}{n} (-1)^n - \frac{\pi}{n} (-1)^n \right] + \frac{1}{n\pi} \left[\frac{1}{n} \sin(nx) \right]_{-\pi}^{\pi} \\ &= \frac{1}{\pi} \left[-\frac{2\pi}{n} (-1)^n \right] + \frac{1}{n^2\pi} [\sin(n\pi) - \sin(-n\pi)] \\ &= -\frac{2}{n} (-1)^n + \frac{1}{n^2\pi} [0 - 0] \\ &= \frac{2}{n} (-1)^{n+1} \end{aligned}$$

Thus, the Fourier series for $f(x) = x$ is:

$$f(x) \sim \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin(nx)$$

2. We use Dirichlet's Convergence Theorem. The function $f(x) = x$ is piecewise smooth on the interval $[-\pi, \pi]$. Its derivative $f'(x) = 1$ is continuous on $(-\pi, \pi)$.

(a) At $x = \pi/2$: Since $f(x)$ is continuous at $x = \pi/2$ (and in the open interval $(-\pi, \pi)$), the Fourier series converges to $f(\pi/2)$.

$$\sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin(n\pi/2) = f(\pi/2) = \pi/2$$

(b) At $x = \pi$: This is an endpoint of the interval. According to the convergence theorem, the Fourier series converges to the average of the left-hand and right-hand limits of the periodic extension of $f(x)$. The periodic extension $f_{ext}(x)$ at $x = \pi$ has a left-hand limit $f(\pi^-) = \pi$. The right-hand limit for the periodic extension at $x = \pi$ is $f(-\pi^+) = -\pi$.

$$\sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin(n\pi) = \frac{f(\pi^-) + f(-\pi^+)}{2} = \frac{\pi + (-\pi)}{2} = 0$$

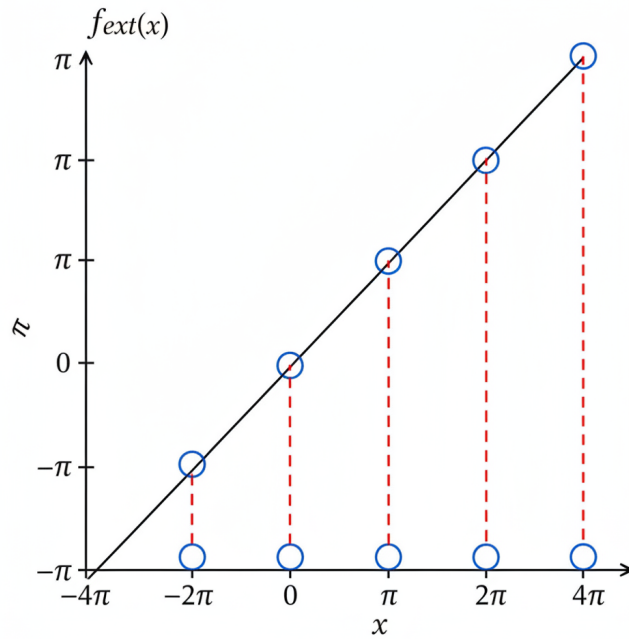
(Note that $\sin(n\pi) = 0$ for all integers n , so the sum is indeed 0).

(c) At $x = -\pi$: Similar to $x = \pi$, this is an endpoint. The left-hand limit for the periodic extension at $x = -\pi$ is $f(\pi^-) = \pi$. The right-hand limit at $x = -\pi$ is $f(-\pi^+) = -\pi$.

$$\sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin(-n\pi) = \frac{f(\pi^-) + f(-\pi^+)}{2} = \frac{\pi + (-\pi)}{2} = 0$$

(Again, $\sin(-n\pi) = 0$, so the sum is 0).

The graph of the function $f(x) = x$ and its periodic extension shows the jumps at integer multiples of π .



Example 1.4.7. Consider the function $f(x)$ defined as:

$$f(x) = \begin{cases} 0 & -\pi < x < 0 \\ 1 & 0 \leq x < \pi \end{cases}$$

1. Find the Fourier series for $f(x)$.
2. Discuss the convergence of the Fourier series at $x = -\pi/2$, $x = 0$, and $x = \pi$.

Solution. 1. The period of the function is $2L = 2\pi$, so $L = \pi$. First, calculate a_0 :

$$\begin{aligned} a_0 &= \frac{1}{L} \int_{-L}^L f(x) dx \\ &= \frac{1}{\pi} \left[\int_{-\pi}^0 0 dx + \int_0^{\pi} 1 dx \right] \\ &= \frac{1}{\pi} [0 + (x)_0^{\pi}] \\ &= \frac{1}{\pi} [\pi - 0] = 1 \end{aligned}$$

Next, calculate a_n :

$$\begin{aligned} a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \\ &= \frac{1}{\pi} \left[\int_{-\pi}^0 0 \cdot \cos(nx) dx + \int_0^{\pi} 1 \cdot \cos(nx) dx \right] \\ &= \frac{1}{\pi} \left[\frac{1}{n} \sin(nx) \right]_0^{\pi} \\ &= \frac{1}{\pi n} [\sin(n\pi) - \sin(0)] \\ &= \frac{1}{\pi n} [0 - 0] = 0 \end{aligned}$$

Now, calculate b_n :

$$\begin{aligned}
 b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \\
 &= \frac{1}{\pi} \left[\int_{-\pi}^0 0 \cdot \sin(nx) dx + \int_0^{\pi} 1 \cdot \sin(nx) dx \right] \\
 &= \frac{1}{\pi} \left[-\frac{1}{n} \cos(nx) \right]_0^{\pi} \\
 &= -\frac{1}{\pi n} [\cos(n\pi) - \cos(0)] \\
 &= -\frac{1}{\pi n} [(-1)^n - 1]
 \end{aligned}$$

If n is even, $(-1)^n = 1$, so $b_n = -\frac{1}{\pi n}[1 - 1] = 0$. If n is odd, $(-1)^n = -1$, so $b_n = -\frac{1}{\pi n}[-1 - 1] = \frac{2}{\pi n}$. So,
 $b_n = \begin{cases} \frac{2}{\pi n} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$. The Fourier series for $f(x)$ is:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

$$f(x) \sim \frac{1}{2} + \sum_{n \text{ odd}}^{\infty} \frac{2}{\pi n} \sin(nx)$$

We can write this by letting $n = 2k - 1$ for $k = 1, 2, \dots$:

$$f(x) \sim \frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{1}{2k-1} \sin((2k-1)x)$$

2. We use Dirichlet's Convergence Theorem. The function $f(x)$ is piecewise smooth. It has a jump discontinuity at $x = 0$. The derivative is $f'(x) = 0$ for $x \in (-\pi, 0)$ and $x \in (0, \pi)$.

- (a) At $x = -\pi/2$: The function is continuous at $x = -\pi/2$, and $f(-\pi/2) = 0$.

$$\frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{1}{2k-1} \sin((2k-1)(-\pi/2)) = f(-\pi/2) = 0$$

- (b) At $x = 0$: This is a point of discontinuity. The Fourier series converges to the average of the left-hand and right-hand limits. $f(0^-) = \lim_{x \rightarrow 0^-} f(x) = 0$. $f(0^+) = \lim_{x \rightarrow 0^+} f(x) = 1$.

$$\frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{1}{2k-1} \sin(0) = \frac{f(0^-) + f(0^+)}{2} = \frac{0+1}{2} = \frac{1}{2}$$

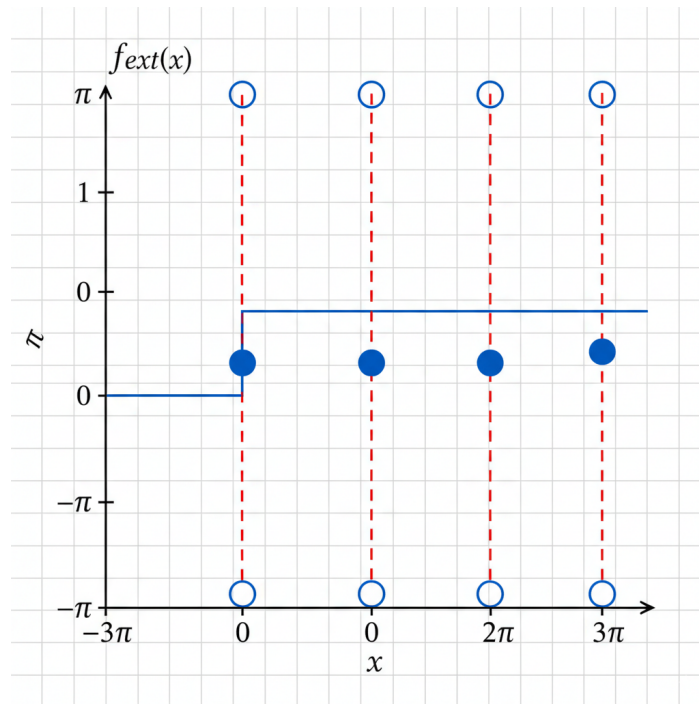
(Note that $\sin(0) = 0$, so the sum evaluates to $1/2$, which is correct).

- (c) At $x = \pi$: This is an endpoint and also a point of discontinuity in the periodic extension. The left-hand limit is $f(\pi^-) = 1$. The right-hand limit for the periodic extension at $x = \pi$ is $f(-\pi^+) = 0$.

$$\frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{1}{2k-1} \sin((2k-1)\pi) = \frac{f(\pi^-) + f(-\pi^+)}{2} = \frac{1+0}{2} = \frac{1}{2}$$

(Note that $\sin((2k-1)\pi) = 0$ for all integers k , so the sum evaluates to $1/2$, which is correct).

The graph of the function $f(x)$ and its periodic extension, showing the jumps at integer multiples of π .



■

Fourier series for sum of infinite series

Fourier series expansions can be used to evaluate the sums of certain infinite series by choosing appropriate values for x in the series representation. Below are two classic examples demonstrating this technique.

Example 1.4.8. We wish to evaluate the series:

$$S_1 = \sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots$$

Consider the even function $f(x) = |x|$ on the interval $[-\pi, \pi]$, extended periodically with period 2π . Its Fourier series expansion is:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx),$$

where the coefficients are:

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x dx = \pi, \quad a_n = \frac{2}{\pi} \int_0^{\pi} x \cos(nx) dx.$$

For $n \geq 1$, integration by parts yields:

$$a_n = \frac{2}{\pi} \left[\frac{x \sin(nx)}{n} + \frac{\cos(nx)}{n^2} \right]_0^{\pi} = \frac{2}{\pi} \cdot \frac{\cos(n\pi) - 1}{n^2} = \frac{2((-1)^n - 1)}{\pi n^2}.$$

Note that $a_n = 0$ for even n , and for odd $n = 2k - 1$:

$$a_{2k-1} = \frac{2((-1)^{2k-1} - 1)}{\pi(2k-1)^2} = -\frac{4}{\pi(2k-1)^2}.$$

Thus, the Fourier series becomes:

$$|x| = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\cos((2k-1)x)}{(2k-1)^2}.$$

Setting $x = 0$ (where $f(0) = 0$):

$$0 = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{1}{(2k-1)^2}.$$

Solving for the series:

$$\sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} = \frac{\pi^2}{8}.$$

Example 1.4.9. We wish to evaluate the alternating series:

$$S_2 = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{2k-1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$$

Consider the odd function $f(x) = x$ on the interval $(-\pi, \pi)$, extended periodically with period 2π . Its Fourier sine series expansion is:

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(nx),$$

where the coefficients are:

$$b_n = \frac{2}{\pi} \int_0^{\pi} x \sin(nx) dx.$$

Integration by parts yields:

$$b_n = \frac{2}{\pi} \left[-\frac{x \cos(nx)}{n} + \frac{\sin(nx)}{n^2} \right]_0^{\pi} = \frac{2}{\pi} \cdot \left(-\frac{\pi \cos(n\pi)}{n} \right) = \frac{2(-1)^{n+1}}{n}.$$

Thus, the Fourier series becomes:

$$x = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(nx).$$

Setting $x = \frac{\pi}{2}$ (where $f(\pi/2) = \pi/2$):

$$\frac{\pi}{2} = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin\left(\frac{n\pi}{2}\right).$$

Note that $\sin(n\pi/2)$ is nonzero only for odd n , with:

$$\sin\left(\frac{n\pi}{2}\right) = \begin{cases} 1 & \text{if } n \equiv 1 \pmod{4}, \\ -1 & \text{if } n \equiv 3 \pmod{4}. \end{cases}$$

Thus, the series becomes:

$$\frac{\pi}{2} = 2 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \right).$$

Solving for the series:

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}.$$

Exercise 1.4.10. Use Fourier series to verify the following sums:

1. Using the Fourier series expansion of $f(x) = x^2$ on $[-\pi, \pi]$, show that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

2. Using the Fourier series expansion of $f(x) = x^2$ on $[-\pi, \pi]$, show that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}.$$

3. Using the Fourier series expansion of $f(x) = x^4$ on $[-\pi, \pi]$, show that

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$

4. Using the Fourier series expansion of $f(x) = |x|$ on $[-\pi, \pi]$, show that

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}.$$

Note: For each problem:

- Find the complete Fourier series expansion of the given function
- Choose an appropriate value of x to evaluate the series
- Justify why the Fourier series converges to the function value at your chosen point
- Manipulate the resulting equation to obtain the desired sum

1.5 Half-Range Fourier Series Expansions

When a function $f(x)$ is defined only on the interval $[0, L]$, we can extend it to the full interval $[-L, L]$ in two natural ways:

1. **Even extension:** Define $f_e(x) = f(-x)$ for $-L \leq x < 0$
2. **Odd extension:** Define $f_o(x) = -f(-x)$ for $-L \leq x < 0$

These extensions lead to Fourier series containing only cosine terms (even extension) or only sine terms (odd extension), known as **half-range expansions**.

1.6 Fourier Cosine Series

For a function $f(x)$ defined on $[0, L]$, its Fourier cosine series is given by:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

where the coefficients are:

$$a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx, \quad n = 0, 1, 2, \dots$$

This series represents the even periodic extension of $f(x)$ with period $2L$.

1.7 Fourier Sine Series

For a function $f(x)$ defined on $[0, L]$, its Fourier sine series is given by:

$$f(x) \sim \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

where the coefficients are:

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx, \quad n = 1, 2, 3, \dots$$

This series represents the odd periodic extension of $f(x)$ with period $2L$.

Example 1.7.1 (Fourier Cosine Series). Find the Fourier cosine series for $f(x) = x$ on $[0, \pi]$.

Solution: We have $L = \pi$. Compute the coefficients:

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x dx = \frac{2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} = \pi$$

For $n \geq 1$:

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \cos(nx) dx$$

Using integration by parts:

$$a_n = \frac{2}{\pi} \left[\frac{x \sin(nx)}{n} + \frac{\cos(nx)}{n^2} \right]_0^{\pi} = \frac{2}{\pi} \left(\frac{\cos(n\pi)}{n^2} - \frac{1}{n^2} \right) = \frac{2[(-1)^n - 1]}{\pi n^2}$$

Thus, the Fourier cosine series is:

$$x \sim \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\cos((2k-1)x)}{(2k-1)^2}$$

Example 1.7.2 (Fourier Sine Series). Find the Fourier sine series for $f(x) = 1$ on $[0, \pi]$.

Solution: We have $L = \pi$. Compute the coefficients:

$$b_n = \frac{2}{\pi} \int_0^{\pi} \sin(nx) dx = \frac{2}{\pi} \left[-\frac{\cos(nx)}{n} \right]_0^{\pi} = \frac{2[1 - (-1)^n]}{\pi n}$$

Thus, the Fourier sine series is:

$$1 \sim \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\sin((2k-1)x)}{2k-1}$$

Example 1.7.3. Find both the cosine and sine series for $f(x) = \pi - x$ on $[0, \pi]$.

Solution for cosine series:

$$a_0 = \frac{2}{\pi} \int_0^\pi (\pi - x) dx = \frac{2}{\pi} \left[\pi x - \frac{x^2}{2} \right]_0^\pi = \pi$$

For $n \geq 1$:

$$a_n = \frac{2}{\pi} \int_0^\pi (\pi - x) \cos(nx) dx = \frac{2}{\pi} \left[\frac{(\pi - x) \sin(nx)}{n} - \frac{\cos(nx)}{n^2} \right]_0^\pi = \frac{2[1 - (-1)^n]}{\pi n^2}$$

Thus, the cosine series is:

$$\pi - x \sim \frac{\pi}{2} + \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\cos((2k-1)x)}{(2k-1)^2}$$

Solution for sine series:

$$b_n = \frac{2}{\pi} \int_0^\pi (\pi - x) \sin(nx) dx = \frac{2}{\pi} \left[-\frac{(\pi - x) \cos(nx)}{n} - \frac{\sin(nx)}{n^2} \right]_0^\pi = \frac{2}{n}$$

Thus, the sine series is:

$$\pi - x \sim 2 \sum_{n=1}^{\infty} \frac{\sin(nx)}{n}$$

1.8 Applications

Half-range expansions are particularly useful in:

- Solving boundary value problems in partial differential equations
- Signal processing where only half-wave symmetry is present
- Problems with specific boundary conditions (e.g., insulated ends)

Exercise 1.8.1. Find the half-range Fourier expansions for the following functions defined on the interval $[0, \pi]$:

1. Find the Fourier cosine series for the function:

$$f(x) = x(\pi - x)$$

2. Find the Fourier sine series for the function:

$$f(x) = \begin{cases} x & \text{for } 0 \leq x \leq \frac{\pi}{2} \\ \pi - x & \text{for } \frac{\pi}{2} < x \leq \pi \end{cases}$$

3. For the function $f(x) = e^x$ on $[0, \pi]$:

- (a) Find the Fourier cosine series expansion
- (b) Find the Fourier sine series expansion

For each problem:

- Compute the appropriate Fourier coefficients (a_n for cosine series, b_n for sine series)
- Write the complete Fourier series expansion
- State the value to which the series converges at points of discontinuity (if any)
- Sketch the periodic extension represented by the Fourier series

1.8.1 Convergence of Coefficients and Parseval's Theorem

A different but related question concerns the convergence of the sequence of the Fourier coefficients themselves. First, we prove Bessel's inequality.

Theorem 1.8.2 (Bessel's Inequality). *Let f be a square-integrable function on $(-\pi, \pi)$ with Fourier coefficients:*

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx.$$

Then the following inequality holds:

$$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \leq \frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 dx.$$

Proof. Consider the N -th partial sum of the Fourier series:

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^N (a_n \cos(nx) + b_n \sin(nx)).$$

Define the mean square error:

$$E_N = \int_{-\pi}^{\pi} [f(x) - S_N(x)]^2 dx.$$

Expanding the square:

$$E_N = \int_{-\pi}^{\pi} [f(x)]^2 dx - 2 \int_{-\pi}^{\pi} f(x) S_N(x) dx + \int_{-\pi}^{\pi} [S_N(x)]^2 dx.$$

We compute each term separately.

First, using the orthogonality of the trigonometric system:

$$\int_{-\pi}^{\pi} S_N(x) f(x) dx = \frac{a_0}{2} \int_{-\pi}^{\pi} f(x) dx + \sum_{n=1}^N \left[a_n \int_{-\pi}^{\pi} f(x) \cos(nx) dx + b_n \int_{-\pi}^{\pi} f(x) \sin(nx) dx \right].$$

Substituting the definitions of the Fourier coefficients:

$$\int_{-\pi}^{\pi} f(x) dx = \pi a_0, \quad \int_{-\pi}^{\pi} f(x) \cos(nx) dx = \pi a_n, \quad \int_{-\pi}^{\pi} f(x) \sin(nx) dx = \pi b_n,$$

we obtain:

$$\int_{-\pi}^{\pi} S_N(x) f(x) dx = \frac{a_0}{2} (\pi a_0) + \sum_{n=1}^N [a_n (\pi a_n) + b_n (\pi b_n)] = \pi \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right).$$

Next, compute the norm of $S_N(x)$:

$$\int_{-\pi}^{\pi} [S_N(x)]^2 dx = \int_{-\pi}^{\pi} \left[\frac{a_0}{2} + \sum_{n=1}^N (a_n \cos(nx) + b_n \sin(nx)) \right]^2 dx.$$

Again using orthogonality:

$$\int_{-\pi}^{\pi} \cos(nx) \cos(mx) dx = \begin{cases} \pi & \text{if } n = m \neq 0, \\ 2\pi & \text{if } n = m = 0, \\ 0 & \text{otherwise,} \end{cases} \quad \int_{-\pi}^{\pi} \sin(nx) \sin(mx) dx = \begin{cases} \pi & \text{if } n = m \neq 0, \\ 0 & \text{otherwise,} \end{cases}$$

and all cross terms vanish. Therefore:

$$\int_{-\pi}^{\pi} [S_N(x)]^2 dx = \pi \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right).$$

Substituting these into the expression for E_N :

$$E_N = \int_{-\pi}^{\pi} [f(x)]^2 dx - 2\pi \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right) + \pi \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right) = \int_{-\pi}^{\pi} [f(x)]^2 dx - \pi \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right).$$

Since $E_N \geq 0$, we conclude:

$$\pi \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right) \leq \int_{-\pi}^{\pi} [f(x)]^2 dx,$$

or equivalently:

$$\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \leq \frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 dx.$$

This holds for all N , so taking the limit as $N \rightarrow \infty$ yields Bessel's inequality:

$$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \leq \frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 dx.$$

Theorem 1.8.3 (Convergence of Fourier Coefficients). *If a function $f(x)$ is bounded and integrable on the interval $(-\pi, \pi)$ with Fourier coefficients a_n and b_n , then the series formed by the sum of the squares of the coefficients converges. That is,*

$$\sum_{n=1}^{\infty} (a_n^2 + b_n^2) \quad \text{converges.}$$

Proof. By Bessel's inequality, for any positive integer N , the Fourier coefficients of f satisfy:

$$\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \leq \frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 dx.$$

Since f is bounded and integrable on $(-\pi, \pi)$, the function $[f(x)]^2$ is also integrable, so the right-hand side is finite. The partial sums

$$S_N = \sum_{n=1}^N (a_n^2 + b_n^2)$$

form a non-decreasing sequence bounded above by $\frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 dx$. Hence, the series $\sum_{n=1}^{\infty} (a_n^2 + b_n^2)$ converges. $\blacksquare$

Remark 1.8.4 (Riemann-Lebesgue Lemma). A necessary condition for the convergence of the series $\sum (a_n^2 + b_n^2)$ is that the terms must approach zero. This leads to the **Riemann-Lebesgue Lemma**, which states that if $f(x)$ is integrable on an interval, then its Fourier coefficients must tend to zero as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} a_n = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} b_n = 0$$

Theorem 1.8.5 (Riemann-Lebesgue Lemma). *Let f be an absolutely integrable function on the interval $[a, b]$, i.e.,*

$$\int_a^b |f(t)| dt < \infty.$$

Then:

$$\lim_{n \rightarrow \infty} \int_a^b f(t) \sin(nt) dt = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \int_a^b f(t) \cos(nt) dt = 0.$$

In particular, for Fourier coefficients of an integrable periodic function, we have:

$$\lim_{n \rightarrow \infty} a_n = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} b_n = 0.$$

Proof. We prove the result for the sine integral; the cosine case is analogous.

First consider a step function $\phi(t)$ on $[a, b]$, which is constant on subintervals:

$$\phi(t) = c_j \quad \text{for } t \in (t_{j-1}, t_j), \quad j = 1, \dots, m.$$

Then:

$$\int_a^b \phi(t) \sin(nt) dt = \sum_{j=1}^m c_j \int_{t_{j-1}}^{t_j} \sin(nt) dt = \sum_{j=1}^m \frac{c_j}{n} [-\cos(nt)]_{t_{j-1}}^{t_j}.$$

Since $|\cos(nt)| \leq 1$, we have:

$$\left| \int_a^b \phi(t) \sin(nt) dt \right| \leq \sum_{j=1}^m \frac{2|c_j|}{n} \leq \frac{2M}{n} \sum_{j=1}^m (t_j - t_{j-1}) = \frac{2M(b-a)}{n},$$

where $M = \max_j |c_j|$. Clearly, this tends to 0 as $n \rightarrow \infty$.

For any integrable function f and any $\epsilon > 0$, there exists a step function ϕ such that:

$$\int_a^b |f(t) - \phi(t)| dt < \frac{\epsilon}{2}.$$

Then:

$$\left| \int_a^b f(t) \sin(nt) dt \right| \leq \left| \int_a^b [f(t) - \phi(t)] \sin(nt) dt \right| + \left| \int_a^b \phi(t) \sin(nt) dt \right|.$$

The first term is bounded by:

$$\int_a^b |f(t) - \phi(t)| dt < \frac{\epsilon}{2}.$$

For the second term, since ϕ is a step function, there exists n_0 such that for all $n > n_0$:

$$\left| \int_a^b \phi(t) \sin(nt) dt \right| < \frac{\epsilon}{2}.$$

Hence, for $n > n_0$:

$$\left| \int_a^b f(t) \sin(nt) dt \right| < \epsilon,$$

which proves the result.

The application to Fourier coefficients follows by taking:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt) dt, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt,$$

and applying the lemma with $n = n$. ■

A stronger result, which turns Bessel's inequality into an equality, holds for functions satisfying Dirichlet's conditions.

Theorem 1.8.6 (Parseval's Theorem). *Let f be a periodic function with period $2L$ that satisfies Dirichlet's conditions, with Fourier series expansion:*

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right).$$

Then the following identity holds:

$$\frac{1}{L} \int_{-L}^L [f(x)]^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

Proof. Let $S_N(x)$ be the N -th partial sum of the Fourier series:

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^N \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right).$$

Consider the mean square error:

$$E_N = \int_{-L}^L [f(x) - S_N(x)]^2 dx.$$

Expanding the square:

$$E_N = \int_{-L}^L [f(x)]^2 dx - 2 \int_{-L}^L f(x) S_N(x) dx + \int_{-L}^L [S_N(x)]^2 dx.$$

By definition of Fourier coefficients, we have:

$$\int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx = La_n, \quad \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx = Lb_n, \quad \text{and} \quad \int_{-L}^L f(x) dx = La_0,$$

Therefore,

$$\int_{-L}^L f(x) S_N(x) dx = \frac{a_0}{2} (La_0) + \sum_{n=1}^N [a_n (La_n) + b_n (Lb_n)] = L \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right).$$

Third term of E_N : Using the orthogonality relations:

$$\int_{-L}^L \cos\left(\frac{n\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx = \begin{cases} L & \text{if } n = m \neq 0, \\ 2L & \text{if } n = m = 0, \\ 0 & \text{otherwise,} \end{cases}$$

$$\int_{-L}^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx = \begin{cases} L & \text{if } n = m \neq 0, \\ 0 & \text{otherwise,} \end{cases}$$

and all cross terms vanish. Therefore:

$$\int_{-L}^L [S_N(x)]^2 dx = L \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right).$$

Substituting into the expression for E_N :

$$E_N = I_1 - 2L \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right) + L \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right) = I_1 - L \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right).$$

Since $E_N \geq 0$, we obtain Bessel's inequality:

$$L \left(\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \right) \leq \int_{-L}^L [f(x)]^2 dx.$$

Under Dirichlet's conditions, the Fourier series converges in the mean, i.e., $\lim_{N \rightarrow \infty} E_N = 0$. Taking the limit as $N \rightarrow \infty$:

$$\lim_{N \rightarrow \infty} E_N = \int_{-L}^L [f(x)]^2 dx - L \left(\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right) = 0.$$

Thus,

$$\int_{-L}^L [f(x)]^2 dx = L \left(\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right),$$

or equivalently,

$$\frac{1}{L} \int_{-L}^L [f(x)]^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

■

Exercise 1.8.7. Solve the following problems related to Fourier series. Show all your work and justify each step.

1. Determine whether the following functions are even, odd, or neither:

- (a) $f(x) = x^3 \cos(x)$
- (b) $g(x) = e^{-|x|}$
- (c) $h(x) = \sin(x^2)$

2. A square wave function is defined as:

$$f(x) = \begin{cases} 1 & \text{for } 0 < x < \pi \\ -1 & \text{for } -\pi < x < 0 \end{cases}$$

with period 2π . Find its Fourier series representation.

3. A sawtooth wave is given by:

$$f(x) = x \quad \text{for } -\pi < x < \pi$$

with period 2π . Determine its Fourier series expansion.

4. A triangular wave is defined as:

$$f(x) = \begin{cases} \pi + x & \text{for } -\pi \leq x \leq 0 \\ \pi - x & \text{for } 0 \leq x \leq \pi \end{cases}$$

with period 2π . Find its Fourier series.

5. For the function $f(x) = x^2$ on the interval $(-\pi, \pi)$:

- (a) Verify Dirichlet's conditions
- (b) Compute the Fourier coefficients a_n and b_n
- (c) Write the complete Fourier series
- (d) Using Parseval's identity, show that $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$

6. Consider the function $f(x) = e^x$ defined on $[0, \pi]$:

- (a) Find its Fourier cosine series expansion
- (b) Find its Fourier sine series expansion
- (c) Compare the convergence rates of both series

7. For the function $f(x) = \cos(x)$ on $[0, \pi]$:

- (a) Find the half-range sine series
- (b) Find the half-range cosine series
- (c) Explain why the results differ

8. Using the Riemann-Lebesgue lemma, show that for any absolutely integrable function $f(x)$ on $[-\pi, \pi]$, the Fourier coefficients satisfy:

$$\lim_{n \rightarrow \infty} a_n = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} b_n = 0$$

9. Consider the function:

$$f(x) = \begin{cases} x & \text{for } 0 \leq x \leq 1 \\ 2 - x & \text{for } 1 < x \leq 2 \end{cases}$$

with period 2. Find its Fourier series representation.

10. For the function $f(x) = \sin(\pi x)$ on $[-1, 1]$:

- (a) Find its Fourier series expansion
- (b) Verify that the series converges to the function at all points
- (c) Using appropriate x values, find the sums:

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{4n^2 - 1} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{4n^2 - 1}$$

Chapter 2

The Fourier Transform

The Fourier transform is a mathematical operation that transforms a function of time (or space) into a function of frequency. It decomposes a function into its constituent frequencies, similar to how a musical chord can be expressed as the frequencies of its constituent notes.

2.1 Definition and Basic Formulas

2.1.1 Fourier Transform

For a function $f(t)$ that is absolutely integrable on $(-\infty, \infty)$, the Fourier transform is defined as:

$$\mathcal{F}\{f(t)\} = F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$$

where ω is the angular frequency.

Example 2.1.1 (Rectangular Pulse). Find the Fourier transform of the rectangular pulse:

$$f(t) = \begin{cases} 1 & \text{for } |t| < a \\ 0 & \text{for } |t| > a \end{cases}$$

Solution.

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt = \int_{-a}^a e^{-i\omega t} dt = \left[\frac{e^{-i\omega t}}{-i\omega} \right]_{-a}^a = \frac{2 \sin(a\omega)}{\omega}$$

This is the well-known unnormalized sinc function: $F(\omega) = 2a \cdot \text{sinc}(a\omega)$ ■

Result 2.1.2. The Gaussian Integral

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

Solution: Let its value be I .

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx$$

We evaluate it by squaring the integral and converting to polar coordinates.

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy \right) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy$$

In polar coordinates, $x^2 + y^2 = r^2$ and the area element $dx dy = r dr d\theta$. The limits for the entire plane are $r \in [0, \infty)$ and $\theta \in [0, 2\pi]$.

$$\begin{aligned} I^2 &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} (r dr) d\theta \\ &= \int_0^{2\pi} \left[-\frac{1}{2} e^{-r^2} \right]_{r=0}^{r=\infty} d\theta \\ &= \int_0^{2\pi} \left(0 - \left(-\frac{1}{2}\right) \right) d\theta \\ &= \int_0^{2\pi} \frac{1}{2} d\theta = \frac{1}{2} [\theta]_0^{2\pi} = \pi \end{aligned}$$

Since $I^2 = \pi$ and the integrand e^{-x^2} is always positive, we take the positive square root.

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

Example 2.1.3 (Gaussian Function). Find the Fourier transform of $f(t) = e^{-at^2}$.

Solution.

$$F(\omega) = \int_{-\infty}^{\infty} e^{-at^2} e^{-i\omega t} dt = \int_{-\infty}^{\infty} e^{-a(t^2 + \frac{i\omega}{a}t)} dt$$

Complete the square:

$$t^2 + \frac{i\omega}{a}t = \left(t + \frac{i\omega}{2a}\right)^2 + \frac{\omega^2}{4a^2}$$

Thus,

$$F(\omega) = e^{-\frac{\omega^2}{4a}} \int_{-\infty}^{\infty} e^{-a(t + \frac{i\omega}{2a})^2} dt = \sqrt{\frac{\pi}{a}} e^{-\frac{\omega^2}{4a}}$$

Example 2.1.4 (Exponential Decay). Find the Fourier transform of $f(t) = e^{-a|t|}$.

Solution.

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} e^{-a|t|} e^{-i\omega t} dt = \int_{-\infty}^0 e^{at} e^{-i\omega t} dt + \int_0^{\infty} e^{-at} e^{-i\omega t} dt \\ &= \int_{-\infty}^0 e^{(a-i\omega)t} dt + \int_0^{\infty} e^{-(a+i\omega)t} dt = \frac{1}{a-i\omega} + \frac{1}{a+i\omega} = \frac{2a}{a^2 + \omega^2} \end{aligned}$$

Example 2.1.5. Consider the rectangular pulse function defined as:

$$f(t) = \begin{cases} 1 & \text{for } -\frac{1}{2} \leq t \leq \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

The Fourier transform of $f(t)$ is given by:

$$F(\omega) = \mathcal{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

Since $f(t)$ is zero outside the interval $[-\frac{1}{2}, \frac{1}{2}]$, we can write:

$$F(\omega) = \int_{-1/2}^{1/2} e^{-i\omega t} dt$$

Evaluating this integral:

$$F(\omega) = \left[\frac{e^{-i\omega t}}{-i\omega} \right]_{-1/2}^{1/2} = \frac{1}{-i\omega} (e^{-i\omega/2} - e^{i\omega/2})$$

Using Euler's formula, we can express this as:

$$F(\omega) = \frac{1}{-i\omega} (-2i \sin(\omega/2)) = \frac{2}{\omega} \sin\left(\frac{\omega}{2}\right)$$

This can be written in terms of the sinc function:

$$F(\omega) = \frac{\sin(\omega/2)}{\omega/2} = \text{sinc}\left(\frac{\omega}{2}\right)$$

where the sinc function is defined as:

$$\text{sinc}(x) = \frac{\sin x}{x}$$

Thus, the Fourier transform of the rectangular pulse is:

$$\boxed{\mathcal{F}\{\text{rect}(t)\} = \text{sinc}\left(\frac{\omega}{2}\right)}$$

Definition 2.1.6 (Inverse Fourier Transform). The inverse Fourier transform is given by:

$$\mathcal{F}^{-1}\{F(\omega)\} = f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$$

2.1.2 Fourier Sine and Cosine Transforms

For real-valued functions, we often use sine and cosine transforms:

Fourier Cosine Transform:

$$\mathcal{F}_c\{f(t)\} = F_c(\omega) = \int_0^\infty f(t) \cos(\omega t) dt$$

Fourier Sine Transform:

$$\mathcal{F}_s\{f(t)\} = F_s(\omega) = \int_0^\infty f(t) \sin(\omega t) dt$$

2.2 Properties of Fourier Transform

1. **Linearity:** $\mathcal{F}\{af(t) + bg(t)\} = aF(\omega) + bG(\omega)$
2. **Time Scaling:** $\mathcal{F}\{f(at)\} = \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$
3. **Time Shifting:** $\mathcal{F}\{f(t - t_0)\} = e^{-i\omega t_0} F(\omega)$
4. **Frequency Shifting:** $\mathcal{F}\{e^{i\omega_0 t} f(t)\} = F(\omega - \omega_0)$
5. **Symmetry:** If $F(\omega)$ is the Fourier transform of $f(t)$, then $\mathcal{F}\{F(t)\} = 2\pi f(-\omega)$
6. **Derivative:** $\mathcal{F}\{f'(t)\} = i\omega F(\omega)$
7. **Convolution Theorem:** $\mathcal{F}\{f(t) * g(t)\} = F(\omega)G(\omega)$

Example 2.2.1. Let $f(t) = e^{-t^2}$ (Gaussian function). We know:

$$\mathcal{F}\{e^{-t^2}\} = \sqrt{\pi} e^{-\omega^2/4}$$

For $a = 2$, we have:

$$\mathcal{F}\{e^{-(2t)^2}\} = \mathcal{F}\{e^{-4t^2}\} = \frac{1}{2} \sqrt{\pi} e^{-(\omega/2)^2/4} = \frac{\sqrt{\pi}}{2} e^{-\omega^2/16}$$

Example 2.2.2. Let $f(t) = \text{rect}(t)$ (rectangular pulse from $-1/2$ to $1/2$). We know:

$$\mathcal{F}\{\text{rect}(t)\} = \text{sinc}(\omega/2\pi)$$

For $t_0 = 2$, we have:

$$\mathcal{F}\{\text{rect}(t - 2)\} = e^{-i2\omega} \text{sinc}(\omega/2\pi)$$

The magnitude remains unchanged while the phase is shifted by -2ω .

Example 2.2.3. Let $f(t) = \text{rect}(t)$ and $\omega_0 = 4\pi$. Then:

$$\mathcal{F}\{e^{i4\pi t} \text{rect}(t)\} = \text{sinc}((\omega - 4\pi)/2\pi)$$

This represents a frequency-shifted version of the sinc function.

Example 2.2.4. Let $f(t) = e^{-|t|}$. We know:

$$\mathcal{F}\{e^{-|t|}\} = \frac{2}{1 + \omega^2}$$

Then:

$$\mathcal{F}\left\{\frac{2}{1 + t^2}\right\} = 2\pi e^{-|\omega|} = 2\pi e^{-|\omega|}$$

Example 2.2.5. Let $f(t) = e^{-t}u(t)$. Then $f'(t) = -e^{-t}u(t) + \delta(t)$. We know:

$$\mathcal{F}\{e^{-t}u(t)\} = \frac{1}{1 + i\omega}$$

Using the derivative property:

$$\mathcal{F}\{f'(t)\} = i\omega \cdot \frac{1}{1 + i\omega} = \frac{i\omega}{1 + i\omega}$$

Example 2.2.6. $\mathcal{F}\{f(t) * g(t)\} = F(\omega)G(\omega)$

Example: Let $f(t) = g(t) = \text{rect}(t)$. The convolution of two rectangular pulses is a triangular pulse:

$$(f * g)(t) = \begin{cases} 1 - |t| & |t| < 1 \\ 0 & |t| \geq 1 \end{cases}$$

We know:

$$\mathcal{F}\{\text{rect}(t)\} = \text{sinc}(\omega/2\pi)$$

By the convolution theorem:

$$\mathcal{F}\{f(t) * g(t)\} = [\text{sinc}(\omega/2\pi)]^2$$

Now, we prove these properties of Fourier transform one by one.

Theorem 2.2.7 (Linearity). *For any constants a and b , and functions $f(t)$ and $g(t)$ with Fourier transforms $F(\omega)$ and $G(\omega)$ respectively:*

$$\mathcal{F}\{af(t) + bg(t)\} = aF(\omega) + bG(\omega)$$

Proof. By the definition of the Fourier transform:

$$\mathcal{F}\{af(t) + bg(t)\} = \int_{-\infty}^{\infty} [af(t) + bg(t)]e^{-i\omega t} dt$$

Using the linearity of integration:

$$= a \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt + b \int_{-\infty}^{\infty} g(t)e^{-i\omega t} dt = aF(\omega) + bG(\omega)$$

■

Theorem 2.2.8 (Time Scaling). *For a constant $a \neq 0$:*

$$\mathcal{F}\{f(at)\} = \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$$

Proof. Consider first the case $a > 0$. Let $u = at$, then $du = a dt$, $dt = du/a$:

$$\mathcal{F}\{f(at)\} = \int_{-\infty}^{\infty} f(at)e^{-i\omega t} dt = \int_{-\infty}^{\infty} f(u)e^{-i\omega(u/a)} \frac{du}{a} = \frac{1}{a} \int_{-\infty}^{\infty} f(u)e^{-i(\omega/a)u} du = \frac{1}{a} F\left(\frac{\omega}{a}\right)$$

For $a < 0$, let $a = -|a|$. Then $u = at = -|a|t$, $du = -|a|dt$, $dt = -du/|a|$:

$$\mathcal{F}\{f(at)\} = \int_{-\infty}^{\infty} f(at)e^{-i\omega t} dt = \int_{\infty}^{-\infty} f(u)e^{-i\omega(u/a)} \left(-\frac{du}{|a|}\right) = \frac{1}{|a|} \int_{-\infty}^{\infty} f(u)e^{-i(\omega/a)u} du = \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$$

Combining both cases gives the result:

$$\mathcal{F}\{f(at)\} = \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$$

■

Theorem 2.2.9 (Time Shifting). *For any real constant t_0 :*

$$\mathcal{F}\{f(t - t_0)\} = e^{-i\omega t_0} F(\omega)$$

Proof. Let $u = t - t_0$, then $t = u + t_0$, $dt = du$:

$$\begin{aligned} \mathcal{F}\{f(t - t_0)\} &= \int_{-\infty}^{\infty} f(t - t_0)e^{-i\omega t} dt = \int_{-\infty}^{\infty} f(u)e^{-i\omega(u+t_0)} du \\ &= e^{-i\omega t_0} \int_{-\infty}^{\infty} f(u)e^{-i\omega u} du = e^{-i\omega t_0} F(\omega) \end{aligned}$$

■

Theorem 2.2.10 (Frequency Shifting). *For any real constant ω_0 :*

$$\mathcal{F}\{e^{i\omega_0 t} f(t)\} = F(\omega - \omega_0)$$

Proof.

$$\mathcal{F}\{e^{i\omega_0 t} f(t)\} = \int_{-\infty}^{\infty} e^{i\omega_0 t} f(t)e^{-i\omega t} dt = \int_{-\infty}^{\infty} f(t)e^{-i(\omega - \omega_0)t} dt = F(\omega - \omega_0)$$

■

Theorem 2.2.11 (Symmetry Property of Fourier Transform). *Let $f(t)$ be a function with Fourier transform $F(\omega)$, denoted by $\mathcal{F}[f(t)] = F(\omega)$. Then the Fourier transform of the function $F(t)$ is given by:*

$$\mathcal{F}[F(t)] = 2\pi f(-\omega)$$

Proof. We begin with the definition of the Inverse Fourier Transform:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$$

Multiplying by 2π gives:

$$2\pi f(t) = \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$$

To obtain the desired form, we replace the variable t with $-\omega$:

$$2\pi f(-\omega) = \int_{-\infty}^{\infty} F(t') e^{-it'\omega} dt'$$

where we have renamed the variable of integration from ω to a dummy variable t' to avoid confusion.

The integral on the right-hand side is, by definition, the Fourier transform of the function $F(t')$.

$$\int_{-\infty}^{\infty} F(t') e^{-i\omega t'} dt' = \mathcal{F}[F(t)]$$

Therefore, by equating the expressions, we have:

$$\mathcal{F}[F(t)] = 2\pi f(-\omega)$$

This completes the proof. ■

Theorem 2.2.12 (Symmetry). *If $F(\omega)$ is the Fourier transform of $f(t)$, then:*

$$\mathcal{F}\{F(t)\} = 2\pi f(-\omega)$$

Proof. By definition of the Fourier transform and its inverse:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$$

Replace t with $-t$:

$$f(-t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{-i\omega t} d\omega$$

Now, by definition of the Fourier transform:

$$\mathcal{F}\{F(t)\} = \int_{-\infty}^{\infty} F(t) e^{-i\omega t} dt$$

Comparing these two expressions, we see that:

$$\mathcal{F}\{F(t)\} = 2\pi f(-\omega)$$

This completes the proof. ■

Theorem 2.2.13 (Derivative). *If $f(t)$ has Fourier transform $F(\omega)$, then:*

$$\mathcal{F}\{f'(t)\} = i\omega F(\omega)$$

Proof. By definition:

$$\mathcal{F}\{f'(t)\} = \int_{-\infty}^{\infty} f'(t) e^{-i\omega t} dt$$

Using integration by parts with:

$$\begin{aligned} u &= e^{-i\omega t} & dv &= f'(t) dt \\ du &= -i\omega e^{-i\omega t} dt & v &= f(t) \end{aligned}$$

we get:

$$\mathcal{F}\{f'(t)\} = [f(t) e^{-i\omega t}]_{-\infty}^{\infty} + i\omega \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

Since $f(t)$ must vanish at $\pm\infty$ for the Fourier transform to exist, the boundary term is zero. Thus:

$$\mathcal{F}\{f'(t)\} = i\omega \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt = i\omega F(\omega)$$

This completes the proof. ■

Remark 2.2.14 (The general Case). If $\mathcal{F}[f(t)] = F(\omega)$, then $\mathcal{F}^n[f(t)] = (i\omega)^n F(\omega)$, where $f(x)$ and all its derivatives vanish at ∞ .

Theorem 2.2.15 (Convolution Theorem). If $f(t)$ and $g(t)$ have Fourier transforms $F(\omega)$ and $G(\omega)$ respectively, then:

$$\mathcal{F}\{f(t) * g(t)\} = F(\omega)G(\omega)$$

where the convolution is defined as:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau)d\tau$$

Proof. By definition:

$$\mathcal{F}\{f(t) * g(t)\} = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(\tau)g(t - \tau)d\tau \right] e^{-i\omega t} dt$$

Interchanging the order of integration (justified by Fubini's theorem):

$$= \int_{-\infty}^{\infty} f(\tau) \left[\int_{-\infty}^{\infty} g(t - \tau)e^{-i\omega t} dt \right] d\tau$$

Make the substitution $u = t - \tau$ in the inner integral:

$$\begin{aligned} &= \int_{-\infty}^{\infty} f(\tau) \left[\int_{-\infty}^{\infty} g(u)e^{-i\omega(u+\tau)} du \right] d\tau \\ &= \int_{-\infty}^{\infty} f(\tau)e^{-i\omega\tau} d\tau \cdot \int_{-\infty}^{\infty} g(u)e^{-i\omega u} du = F(\omega)G(\omega) \end{aligned}$$

This completes the proof. ■

Theorem 2.2.16 (Modulation Property). If $F(\omega)$ is the Fourier transform of $f(t)$, then for any real constant ω_0 :

$$(i) \quad \mathcal{F}\{f(t) \cos(\omega_0 t)\} = \frac{1}{2}[F(\omega + \omega_0) + F(\omega - \omega_0)]$$

$$(ii) \quad \mathcal{F}\{f(t) \sin(\omega_0 t)\} = \frac{i}{2}[F(\omega + \omega_0) - F(\omega - \omega_0)]$$

Proof. (i) We begin by expressing the cosine function in terms of complex exponentials using Euler's formula:

$$\cos(\omega_0 t) = \frac{e^{i\omega_0 t} + e^{-i\omega_0 t}}{2}$$

Now, applying the Fourier transform to the modulated function:

$$\begin{aligned} \mathcal{F}\{f(t) \cos(\omega_0 t)\} &= \int_{-\infty}^{\infty} f(t) \cos(\omega_0 t) e^{-i\omega t} dt \\ &= \int_{-\infty}^{\infty} f(t) \left(\frac{e^{i\omega_0 t} + e^{-i\omega_0 t}}{2} \right) e^{-i\omega t} dt \\ &= \frac{1}{2} \int_{-\infty}^{\infty} f(t) e^{i\omega_0 t} e^{-i\omega t} dt + \frac{1}{2} \int_{-\infty}^{\infty} f(t) e^{-i\omega_0 t} e^{-i\omega t} dt \\ &= \frac{1}{2} \int_{-\infty}^{\infty} f(t) e^{-i(\omega - \omega_0)t} dt + \frac{1}{2} \int_{-\infty}^{\infty} f(t) e^{-i(\omega + \omega_0)t} dt \\ &= \frac{1}{2} F(\omega - \omega_0) + \frac{1}{2} F(\omega + \omega_0) \end{aligned}$$
■

Definition 2.2.17. The unit step function $u_0(t)$ is defined as

$$u_0(t) = \begin{cases} 0; & t < 0 \\ 1; & t > 0 \end{cases}$$

Example 2.2.18. Find the solution of the differential equation

$$y' - 2y = u_0(t)e^{-2t}, -\infty < t < \infty.$$

using the Fourier transform.

Solution. Let $\hat{y}(\omega) = \mathcal{F}\{y(t)\}(\omega)$. Using the property $\mathcal{F}\{y'(t)\}(\omega) = i\omega\hat{y}(\omega)$, we obtain:

$$(i\omega - 2)\hat{y}(\omega) = \mathcal{F}\{u_0(t)e^{-2t}\}(\omega). \quad (1)$$

Now,

$$\mathcal{F}\{u_0(t)e^{-2t}\}(\omega) = \int_0^\infty e^{-(2+i\omega)t} dt = \frac{1}{2+i\omega}.$$

Therefore,

$$(i\omega - 2)\hat{y}(\omega) = \frac{1}{2+i\omega} \Rightarrow \hat{y}(\omega) = \frac{1}{(i\omega - 2)(2+i\omega)}.$$

Note that $2+i\omega = i\omega+2$, so:

$$\hat{y}(\omega) = \frac{1}{(i\omega - 2)(i\omega + 2)} = \frac{1}{(i\omega)^2 - 4} = -\frac{1}{\omega^2 + 4}.$$

We know that,

$$\mathcal{F}\left\{e^{-2|t|}\right\}(\omega) = \frac{4}{\omega^2 + 4} \Rightarrow \mathcal{F}\left\{\frac{1}{4}e^{-2|t|}\right\}(\omega) = \frac{1}{\omega^2 + 4}.$$

Thus,

$$\hat{y}(\omega) = -\frac{1}{\omega^2 + 4} = -\mathcal{F}\left\{\frac{1}{4}e^{-2|t|}\right\}(\omega) = \mathcal{F}\left\{-\frac{1}{4}e^{-2|t|}\right\}(\omega),$$

so the inverse transform gives:

$$y(t) = -\frac{1}{4}e^{-2|t|}.$$

Verification

- For $t > 0$: $y(t) = -\frac{1}{4}e^{-2t}$, $y'(t) = \frac{1}{2}e^{-2t}$, and

$$y' - 2y = \frac{1}{2}e^{-2t} + \frac{1}{2}e^{-2t} = e^{-2t}.$$

- For $t < 0$: $y(t) = -\frac{1}{4}e^{2t}$, $y'(t) = -\frac{1}{2}e^{2t}$, and

$$y' - 2y = -\frac{1}{2}e^{2t} + \frac{1}{2}e^{2t} = 0.$$

The solution satisfies the equation for all t .

Theorem 2.2.19. Let $f(t)$ be piecewise continuous on every interval $[-l, l]$. Suppose that

$$\int_{-\infty}^{\infty} |t^n f(t)| dt$$

converges. Then

$$\mathcal{F}[t^n f(t)](\omega) = i^n \frac{d^n}{d\omega^n} F(\omega),$$

where $F(\omega) = \mathcal{F}[f(t)](\omega)$ is the Fourier transform of $f(t)$.

Proof. We prove the result by induction on n .

For $n = 0$, the statement is trivial since $\mathcal{F}[f(t)](\omega) = F(\omega) = i^0 F^{(0)}(\omega)$.

Now, for $n = 1$, we wish to show that

$$\mathcal{F}[tf(t)](\omega) = i \frac{d}{d\omega} F(\omega).$$

Recall that

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt.$$

Under the assumption that $\int_{-\infty}^{\infty} |tf(t)| dt$ converges, we may differentiate under the integral sign. Specifically, consider the derivative with respect to ω :

$$\frac{d}{d\omega} F(\omega) = \frac{d}{d\omega} \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt.$$

Note that

$$\frac{\partial}{\partial \omega} (f(t)e^{-i\omega t}) = -itf(t)e^{-i\omega t},$$

and

$$|-itf(t)e^{-i\omega t}| = |tf(t)|.$$

Since $\int_{-\infty}^{\infty} |tf(t)|dt$ converges, the dominated convergence theorem (or Leibniz's rule for differentiation under the integral sign) applies, and we have

$$\frac{d}{d\omega} F(\omega) = \int_{-\infty}^{\infty} \frac{\partial}{\partial \omega} (f(t)e^{-i\omega t}) dt = \int_{-\infty}^{\infty} -itf(t)e^{-i\omega t} dt = -i \int_{-\infty}^{\infty} tf(t)e^{-i\omega t} dt.$$

Therefore,

$$\int_{-\infty}^{\infty} tf(t)e^{-i\omega t} dt = i \frac{d}{d\omega} F(\omega),$$

which is exactly

$$\mathcal{F}[tf(t)](\omega) = iF'(\omega).$$

Now assume that the result holds for $n-1$, i.e.,

$$\mathcal{F}[t^{n-1}f(t)](\omega) = i^{n-1}F^{(n-1)}(\omega).$$

Define $g(t) = t^{n-1}f(t)$. Then $t^n f(t) = tg(t)$. Note that $g(t)$ is piecewise continuous on every finite interval, and by hypothesis,

$$\int_{-\infty}^{\infty} |tg(t)|dt = \int_{-\infty}^{\infty} |t^n f(t)|dt$$

converges. Therefore, we may apply the $n=1$ case to $g(t)$ to obtain

$$\mathcal{F}[tg(t)](\omega) = i \frac{d}{d\omega} \mathcal{F}[g(t)](\omega).$$

But by the induction hypothesis,

$$\mathcal{F}[g(t)](\omega) = i^{n-1}F^{(n-1)}(\omega),$$

so

$$\mathcal{F}[t^n f(t)](\omega) = i \frac{d}{d\omega} (i^{n-1}F^{(n-1)}(\omega)) = i^n F^{(n)}(\omega).$$

This completes the induction step, and the theorem is proved. ■

Theorem 2.2.20. *Let $f(t)$ be piecewise continuous on every interval $[-l, l]$. Suppose that*

$$\int_{-\infty}^{\infty} |f(t)| dt$$

converges. Let $\mathcal{F}[f(t)] = F(\omega)$ and suppose that $F(0) = 0$. Then

$$\mathcal{F}\left[\int_{-\infty}^t f(\tau) d\tau\right] = \frac{1}{i\omega} F(\omega).$$

Proof. Let the function $g(t)$ be defined as the integral of $f(t)$:

$$g(t) = \int_{-\infty}^t f(\tau) d\tau$$

Our goal is to compute the Fourier transform of $g(t)$, which we denote by $G(\omega)$. By the definition of the Fourier transform, we have:

$$G(\omega) = \mathcal{F}[g(t)] = \int_{-\infty}^{\infty} g(t)e^{-i\omega t} dt = \int_{-\infty}^{\infty} \left(\int_{-\infty}^t f(\tau) d\tau \right) e^{-i\omega t} dt$$

We can solve this integral using integration by parts, where $\int u dv = uv - \int v du$. Let's choose our parts as follows:

$$\begin{aligned} u &= \int_{-\infty}^t f(\tau) d\tau & dv &= e^{-i\omega t} dt \\ du &= f(t) dt & v &= \frac{e^{-i\omega t}}{-i\omega} \end{aligned}$$

Applying the integration by parts formula gives:

$$G(\omega) = \left[\left(\int_{-\infty}^t f(\tau) d\tau \right) \frac{e^{-i\omega t}}{-i\omega} \right]_{t=-\infty}^{t=\infty} - \int_{-\infty}^{\infty} \left(\frac{e^{-i\omega t}}{-i\omega} \right) f(t) dt$$

Now, we must evaluate the boundary term. Let's analyze it at the limits $t \rightarrow \infty$ and $t \rightarrow -\infty$.

1. **At the upper limit** ($t \rightarrow \infty$): The term is $\lim_{t \rightarrow \infty} \left(\int_{-\infty}^t f(\tau) d\tau \right) \frac{e^{-i\omega t}}{-i\omega}$. The integral part becomes:

$$\lim_{t \rightarrow \infty} \int_{-\infty}^t f(\tau) d\tau = \int_{-\infty}^{\infty} f(\tau) d\tau$$

We can relate this integral to the Fourier transform $F(\omega)$ evaluated at $\omega = 0$:

$$F(0) = \int_{-\infty}^{\infty} f(t) e^{-i(0)t} dt = \int_{-\infty}^{\infty} f(t) dt$$

The theorem gives the crucial condition that $F(0) = 0$. Therefore, $\int_{-\infty}^{\infty} f(\tau) d\tau = 0$. Since the integral part of the boundary term goes to 0 and the term $\frac{e^{-i\omega t}}{-i\omega}$ is bounded, the entire expression evaluates to 0 at the upper limit.

2. **At the lower limit** ($t \rightarrow -\infty$): The term is $\lim_{t \rightarrow -\infty} \left(\int_{-\infty}^t f(\tau) d\tau \right) \frac{e^{-i\omega t}}{-i\omega}$. The integral part becomes:

$$\lim_{t \rightarrow -\infty} \int_{-\infty}^t f(\tau) d\tau = 0$$

by the definition of an improper integral converging. Thus, the boundary term is also 0 at the lower limit.

Since the boundary term vanishes completely, we are left with only the integral part from our integration by parts formula:

$$\begin{aligned} G(\omega) &= - \int_{-\infty}^{\infty} \left(\frac{e^{-i\omega t}}{-i\omega} \right) f(t) dt \\ &= \frac{1}{i\omega} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \end{aligned}$$

The remaining integral is, by definition, the Fourier transform of $f(t)$, which is $F(\omega)$. Therefore, we have shown that:

$$G(\omega) = \mathcal{F} \left[\int_{-\infty}^t f(\tau) d\tau \right] = \frac{1}{i\omega} F(\omega)$$

This completes the proof. ■

Example 2.2.21. Using the convolution theorem, find the inverse Fourier transform of the function

$$Y(\omega) = \frac{1}{12 + 7i\omega - \omega^2}$$

That is, find $y(t) = \mathcal{F}^{-1}[Y(\omega)]$.

Solution. The strategy is to use the convolution theorem in its "inverse" form:

$$\mathcal{F}^{-1}[F(\omega)G(\omega)] = (f * g)(t)$$

where $f(t) = \mathcal{F}^{-1}[F(\omega)]$ and $g(t) = \mathcal{F}^{-1}[G(\omega)]$. First, we must decompose the given function $Y(\omega)$ into a product of two simpler functions, $F(\omega)$ and $G(\omega)$. We do this by factoring the denominator.

Step 1: Factor the Denominator Let's rearrange the denominator and factor it. Notice that $-\omega^2 = (i\omega)^2$.

$$\begin{aligned} 12 + 7i\omega - \omega^2 &= 12 + 7(i\omega) + (i\omega)^2 \\ &= ((i\omega) + 3)((i\omega) + 4) \\ &= (3 + i\omega)(4 + i\omega) \end{aligned}$$

So, we can write $Y(\omega)$ as a product:

$$Y(\omega) = \frac{1}{(3 + i\omega)} \cdot \frac{1}{(4 + i\omega)}$$

Step 2: Identify the Individual Transforms and their Inverses We can now identify our two functions in the frequency domain:

$$F(\omega) = \frac{1}{3 + i\omega} \quad \text{and} \quad G(\omega) = \frac{1}{4 + i\omega}$$

We use the standard Fourier transform pair for a one-sided decaying exponential:

$$\mathcal{F}^{-1} \left[\frac{1}{a + i\omega} \right] = e^{-at} u(t)$$

where $u(t)$ is the Heaviside unit step function. Applying this rule to our functions, we find their corresponding time-domain functions:

$$\begin{aligned} f(t) &= \mathcal{F}^{-1}[F(\omega)] = e^{-3t}u(t) \\ g(t) &= \mathcal{F}^{-1}[G(\omega)] = e^{-4t}u(t) \end{aligned}$$

Step 3: Compute the Convolution According to the convolution theorem, the desired function $y(t)$ is the convolution of $f(t)$ and $g(t)$.

$$y(t) = (f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau) d\tau$$

Substituting our functions:

$$y(t) = \int_{-\infty}^{\infty} [e^{-3\tau}u(\tau)] \cdot [e^{-4(t-\tau)}u(t - \tau)] d\tau$$

The Heaviside functions $u(\tau)$ and $u(t - \tau)$ constrain the limits of integration. The integrand is non-zero only when both $\tau \geq 0$ and $t - \tau \geq 0$ (which implies $\tau \leq t$). Therefore, the integration limits become 0 to t . This also implies that the result is zero for $t < 0$. For $t \geq 0$:

$$\begin{aligned} y(t) &= \int_0^t e^{-3\tau} e^{-4(t-\tau)} d\tau \\ &= \int_0^t e^{-3\tau} e^{-4t} e^{4\tau} d\tau \\ &= e^{-4t} \int_0^t e^{\tau} d\tau \\ &= e^{-4t} [e^{\tau}]_0^t \\ &= e^{-4t} (e^t - e^0) \\ &= e^{-4t} (e^t - 1) \\ &= e^{-3t} - e^{-4t} \end{aligned}$$

Since the result is valid only for $t \geq 0$, we multiply by the Heaviside function to get the final answer.

$$y(t) = (e^{-3t} - e^{-4t})u(t)$$

■

Example 2.2.22. Solve $\frac{dy}{dx} + y = e^{-|x|}$ using the Fourier transform.

Proof. Applying the Fourier transform $\mathcal{F}$ to the equation:

$$\mathcal{F}\left\{\frac{dy}{dx}\right\} + \mathcal{F}\{y\} = \mathcal{F}\{e^{-|x|}\}$$

Using $\mathcal{F}\{y'\}(\omega) = i\omega Y(\omega)$ and $\mathcal{F}\{e^{-|x|}\}(\omega) = \frac{2}{1+\omega^2}$:

$$\begin{aligned} i\omega Y(\omega) + Y(\omega) &= \frac{2}{1+\omega^2} \\ (1+i\omega)Y(\omega) &= \frac{2}{1+\omega^2} \\ Y(\omega) &= \frac{2}{(1+i\omega)(1+\omega^2)} = \frac{2}{(1+i\omega)(1-i\omega)(1+i\omega)} = \frac{2}{(1-i\omega)(1+i\omega)^2} \end{aligned}$$

Perform partial fraction decomposition for $Y(\omega)$:

$$Y(\omega) = \frac{A}{1-i\omega} + \frac{B}{1+i\omega} + \frac{C}{(1+i\omega)^2}$$

Solving for coefficients yields $A = \frac{1}{2}$, $B = \frac{1}{2}$, $C = 1$.

$$Y(\omega) = \frac{1/2}{1-i\omega} + \frac{1/2}{1+i\omega} + \frac{1}{(1+i\omega)^2}$$

Now, apply the inverse Fourier transform $\mathcal{F}^{-1}$. We use standard transform pairs:

- $\mathcal{F}^{-1}\left\{\frac{1}{a-i\omega}\right\}(x) = e^{ax}H(-x)$

- $\mathcal{F}^{-1} \left\{ \frac{1}{a+i\omega} \right\} (x) = e^{-ax} H(x)$
- $\mathcal{F}^{-1} \left\{ \frac{1}{(a+i\omega)^2} \right\} (x) = x e^{-ax} H(x)$

For $a = 1$:

$$y(x) = \frac{1}{2} e^x H(-x) + \frac{1}{2} e^{-x} H(x) + x e^{-x} H(x)$$

Combining terms for $x \geq 0$ and $x < 0$:

$$y(x) = \begin{cases} \frac{1}{2} e^x & \text{if } x < 0 \\ \left(\frac{1}{2} + x\right) e^{-x} & \text{if } x \geq 0 \end{cases}$$

■

- Exercise 2.2.23.**
1. Find the Fourier transform of $f(t) = \delta(t - t_0)$, where δ is the Dirac delta function.
 2. Find the Fourier transform of $f(t) = \cos(\omega_0 t)$.
 3. Using the convolution theorem, find the Fourier transform of the triangular function:

$$f(t) = \begin{cases} 1 - |t|/a & \text{for } |t| < a \\ 0 & \text{for } |t| > a \end{cases}$$

2.3 Discrete Fourier Transform (DFT)

2.3.1 Definition

The Discrete Fourier Transform (DFT) converts a finite sequence of equally-spaced samples of a function into a sequence of coefficients of a finite combination of complex sinusoids.

For a sequence of N complex numbers $x_0, x_1, \dots, x_{N-1}$, the DFT is defined as:

$$X_k = \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi kn/N} \quad \text{for } k = 0, 1, \dots, N-1$$

where:

- X_k is the k -th frequency component
- x_n is the n -th input sample
- N is the total number of samples
- i is the imaginary unit ($i^2 = -1$)

2.3.2 Inverse DFT

The inverse DFT reconstructs the original sequence from the frequency components:

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k \cdot e^{i2\pi kn/N} \quad \text{for } n = 0, 1, \dots, N-1$$

2.3.3 Properties of DFT

1. **Linearity:** For any sequences x_n and y_n and constants a and b :

$$\mathcal{F}\{ax_n + by_n\} = a\mathcal{F}\{x_n\} + b\mathcal{F}\{y_n\}$$

2. **Time Shifting:** Shifting the input sequence by m samples multiplies the DFT by a phase factor:

$$\mathcal{F}\{x_{n-m}\} = X_k \cdot e^{-i2\pi km/N}$$

3. **Frequency Shifting:** Multiplying the input by a complex exponential shifts the frequency components:

$$\mathcal{F}\{x_n \cdot e^{i2\pi mn/N}\} = X_{k-m}$$

4. **Convolution Theorem:** Circular convolution in time domain equals pointwise multiplication in frequency domain:

$$\mathcal{F}\{x_n * y_n\} = X_k \cdot Y_k$$

5. **Parseval's Theorem:** The total energy is conserved between time and frequency domains:

$$\sum_{n=0}^{N-1} |x_n|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X_k|^2$$

6. **Symmetry:** For real-valued input signals:

$$X_k = X_{N-k}^*$$

where $*$ denotes complex conjugation.

2.4 Fast Fourier Transform (FFT)

2.4.1 Introduction

The Fast Fourier Transform (FFT) is an algorithm that efficiently computes the DFT and its inverse. While a direct implementation of the DFT requires $O(N^2)$ operations, the FFT reduces this to $O(N \log N)$ operations.

2.4.2 Cooley-Tukey Algorithm (Radix-2 FFT)

The most common FFT algorithm is the Cooley-Tukey algorithm, which uses a divide-and-conquer approach. It works by recursively breaking down a DFT of size N into smaller DFTs.

$$\begin{aligned} X_k &= \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi kn/N} \\ &= \sum_{m=0}^{N/2-1} x_{2m} \cdot e^{-i2\pi k(2m)/N} + \sum_{m=0}^{N/2-1} x_{2m+1} \cdot e^{-i2\pi k(2m+1)/N} \\ &= \sum_{m=0}^{N/2-1} x_{2m} \cdot e^{-i2\pi km/(N/2)} + e^{-i2\pi k/N} \sum_{m=0}^{N/2-1} x_{2m+1} \cdot e^{-i2\pi km/(N/2)} \end{aligned}$$

This shows that the DFT can be computed as the sum of two smaller DFTs:

- One for the even-indexed elements
- One for the odd-indexed elements (with a twiddle factor)

2.4.3 FFT Algorithm Steps

1. If $N = 1$, return x_0 (base case)
2. Separate even and odd indices:

$$x_{\text{even}} = [x_0, x_2, \dots, x_{N-2}], \quad x_{\text{odd}} = [x_1, x_3, \dots, x_{N-1}]$$

3. Recursively compute DFTs of both halves:

$$X_{\text{even}} = \text{FFT}(x_{\text{even}}), \quad X_{\text{odd}} = \text{FFT}(x_{\text{odd}})$$

4. Combine the results using the twiddle factors $\omega_N^k = e^{-i2\pi k/N}$:

$$X_k = X_{\text{even}}[k] + \omega_N^k \cdot X_{\text{odd}}[k]$$

$$X_{k+N/2} = X_{\text{even}}[k] - \omega_N^k \cdot X_{\text{odd}}[k] \quad \text{for } k = 0, 1, \dots, N/2 - 1$$

2.4.4 Properties and Advantages of FFT

- **Speed:** Reduces computation time from $O(N^2)$ to $O(N \log N)$
- **Memory Efficiency:** Many implementations use in-place computation
- **Numerical Stability:** Generally well-conditioned algorithm
- **Flexibility:** Works for any N , though most efficient when N is a power of 2

Chapter 3

The Laplace Transform

While the Fourier series is an excellent tool for analyzing periodic functions and steady-state phenomena, many problems in engineering and physics involve functions that are not periodic. These often include transient behaviors, such as the response of a circuit immediately after a switch is flipped, or the motion of a damped spring system after an initial push.

The **Laplace Transform** is an integral transform that is particularly powerful for solving linear ordinary differential equations (ODEs), especially initial value problems. Its primary strength lies in its ability to convert a differential equation in the time domain (variable t) into an algebraic equation in a complex frequency domain (variable s). Solving the algebraic equation and then transforming the result back to the time domain is often much simpler than solving the original ODE directly.

3.1 Introduction

Definition 3.1.1 (Laplace Transform). Let $f(t)$ be a function defined for all $t \geq 0$. The **Laplace transform** of $f(t)$, denoted by $\mathcal{L}\{f(t)\}$ or $F(s)$, is defined by the improper integral:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

provided the integral converges. The variable s is a complex variable, $s = \sigma + i\omega$, but for most introductory purposes, we can consider it a real variable. The function $F(s)$ is the transform of $f(t)$.

For the integral to exist, the function $f(t)$ must satisfy certain conditions. Specifically, $f(t)$ must be **piecewise continuous** on $[0, \infty)$ and be of **exponential order**. A function $f(t)$ is of exponential order if there exist constants $M > 0$ and c such that $|f(t)| \leq Me^{ct}$ for all $t \geq 0$. This condition essentially ensures that $f(t)$ does not grow faster than an exponential function, allowing the term e^{-st} to dominate and ensure convergence for a sufficiently large s .

3.1.1 Laplace Transforms of Elementary Functions

We derive the Laplace transforms of several basic functions directly from the definition.

1. The constant function, $f(t) = 1$

$$\begin{aligned} \mathcal{L}\{1\} &= \int_0^{\infty} e^{-st}(1) dt = \left[-\frac{1}{s} e^{-st} \right]_0^{\infty} \\ &= \lim_{b \rightarrow \infty} \left(-\frac{1}{s} e^{-sb} \right) - \left(-\frac{1}{s} e^0 \right) \\ &= 0 - \left(-\frac{1}{s} \right) = \frac{1}{s} \quad (\text{for } s > 0) \end{aligned}$$

2. The exponential function, $f(t) = e^{at}$

$$\begin{aligned} \mathcal{L}\{e^{at}\} &= \int_0^{\infty} e^{-st} e^{at} dt = \int_0^{\infty} e^{-(s-a)t} dt \\ &= \left[-\frac{1}{s-a} e^{-(s-a)t} \right]_0^{\infty} \\ &= 0 - \left(-\frac{1}{s-a} \right) = \frac{1}{s-a} \quad (\text{for } s > a) \end{aligned}$$

3. **The sine and cosine functions, $f(t) = \sin(at)$ and $f(t) = \cos(at)$** A direct integration is possible but tedious. A more elegant method uses Euler's formula, $e^{iat} = \cos(at) + i\sin(at)$, and the known transform of the exponential function.

$$\begin{aligned}\mathcal{L}\{e^{iat}\} &= \frac{1}{s - ia} \\ &= \frac{1}{s - ia} \cdot \frac{s + ia}{s + ia} = \frac{s + ia}{s^2 - (ia)^2} = \frac{s + ia}{s^2 + a^2} \\ &= \frac{s}{s^2 + a^2} + i\frac{a}{s^2 + a^2}\end{aligned}$$

By the linearity of the integral (and thus the transform), we also have:

$$\mathcal{L}\{e^{iat}\} = \mathcal{L}\{\cos(at) + i\sin(at)\} = \mathcal{L}\{\cos(at)\} + i\mathcal{L}\{\sin(at)\}$$

Equating the real and imaginary parts of these two expressions gives:

$$\begin{aligned}\mathcal{L}\{\cos(at)\} &= \frac{s}{s^2 + a^2} \quad (\text{for } s > 0) \\ \mathcal{L}\{\sin(at)\} &= \frac{a}{s^2 + a^2} \quad (\text{for } s > 0)\end{aligned}$$

4. **The power function, $f(t) = t^n$ for integer $n \geq 0$** For $n = 1$, we use integration by parts:

$$\begin{aligned}\mathcal{L}\{t\} &= \int_0^\infty te^{-st} dt \quad (u = t, dv = e^{-st} dt) \\ &= \left[-\frac{t}{s}e^{-st} \right]_0^\infty - \int_0^\infty \left(-\frac{1}{s} \right) e^{-st} dt \\ &= (0 - 0) + \frac{1}{s} \int_0^\infty e^{-st} dt = \frac{1}{s} \mathcal{L}\{1\} = \frac{1}{s} \left(\frac{1}{s} \right) = \frac{1}{s^2} \quad (\text{for } s > 0)\end{aligned}$$

By repeating this process, a general formula can be derived:

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \quad (\text{for } s > 0)$$

3.2 Properties of the Laplace Transform

3.2.1 Linearity

The Laplace transform is a linear operator. This is its most fundamental property and follows directly from the linearity of integration.

Theorem 3.2.1 (Linearity of the Laplace Transform). *If the Laplace transforms of $f_1(t)$ and $f_2(t)$ exist, and c_1 and c_2 are any constants, then the Laplace transform of their linear combination also exists and is given by:*

$$\mathcal{L}\{c_1f_1(t) + c_2f_2(t)\} = c_1\mathcal{L}\{f_1(t)\} + c_2\mathcal{L}\{f_2(t)\} = c_1F_1(s) + c_2F_2(s)$$

Proof. By definition,

$$\begin{aligned}\mathcal{L}\{c_1f_1(t) + c_2f_2(t)\} &= \int_0^\infty e^{-st}[c_1f_1(t) + c_2f_2(t)] dt \\ &= \int_0^\infty (c_1e^{-st}f_1(t) + c_2e^{-st}f_2(t)) dt \\ &= c_1 \int_0^\infty e^{-st}f_1(t) dt + c_2 \int_0^\infty e^{-st}f_2(t) dt \\ &= c_1\mathcal{L}\{f_1(t)\} + c_2\mathcal{L}\{f_2(t)\}\end{aligned}$$

Example 3.2.2 (Using Linearity). Find the Laplace transform of $f(t) = 3 + 4\cos(2t) - 5e^{-t}$.

Solution. Using the linearity property, we can transform each term separately:

$$\begin{aligned}\mathcal{L}\{3 + 4\cos(2t) - 5e^{-t}\} &= 3\mathcal{L}\{1\} + 4\mathcal{L}\{\cos(2t)\} - 5\mathcal{L}\{e^{-t}\} \\ &= 3\left(\frac{1}{s}\right) + 4\left(\frac{s}{s^2 + 2^2}\right) - 5\left(\frac{1}{s - (-1)}\right) \\ &= \frac{3}{s} + \frac{4s}{s^2 + 4} - \frac{5}{s + 1}\end{aligned}$$

Example 3.2.3 (Transforms of Hyperbolic Functions). Find the Laplace transform of $\cosh(at)$ and $\sinh(at)$.

Solution. We use the definitions $\cosh(at) = \frac{e^{at} + e^{-at}}{2}$ and $\sinh(at) = \frac{e^{at} - e^{-at}}{2}$ along with the linearity property.

$$\begin{aligned}\mathcal{L}\{\cosh(at)\} &= \mathcal{L}\left\{\frac{1}{2}e^{at} + \frac{1}{2}e^{-at}\right\} = \frac{1}{2}\mathcal{L}\{e^{at}\} + \frac{1}{2}\mathcal{L}\{e^{-at}\} \\ &= \frac{1}{2}\left(\frac{1}{s-a}\right) + \frac{1}{2}\left(\frac{1}{s+a}\right) = \frac{1}{2}\frac{(s+a) + (s-a)}{(s-a)(s+a)} \\ &= \frac{1}{2}\frac{2s}{s^2 - a^2} = \frac{s}{s^2 - a^2} \quad (\text{for } s > |a|)\end{aligned}$$

Similarly for $\sinh(at)$:

$$\begin{aligned}\mathcal{L}\{\sinh(at)\} &= \mathcal{L}\left\{\frac{1}{2}e^{at} - \frac{1}{2}e^{-at}\right\} = \frac{1}{2}\mathcal{L}\{e^{at}\} - \frac{1}{2}\mathcal{L}\{e^{-at}\} \\ &= \frac{1}{2}\left(\frac{1}{s-a}\right) - \frac{1}{2}\left(\frac{1}{s+a}\right) = \frac{1}{2}\frac{(s+a) - (s-a)}{(s-a)(s+a)} \\ &= \frac{1}{2}\frac{2a}{s^2 - a^2} = \frac{a}{s^2 - a^2} \quad (\text{for } s > |a|)\end{aligned}$$

■

3.2.2 First Translation Theorem (s-Shifting)

This property relates the multiplication of a function by an exponential in the time domain to a shift in the frequency domain. It is extremely useful for finding transforms of functions like $e^{at}\cos(bt)$.

Theorem 3.2.4 (First Translation or s-Shifting Theorem). *If $\mathcal{L}\{f(t)\} = F(s)$, then for any real or complex number a :*

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$$

Proof. The proof follows directly from the definition of the Laplace transform:

$$\begin{aligned}\mathcal{L}\{e^{at}f(t)\} &= \int_0^\infty e^{-st}(e^{at}f(t))dt \\ &= \int_0^\infty e^{-(s-a)t}f(t)dt\end{aligned}$$

This last integral is precisely the definition of the Laplace transform of $f(t)$, but with s replaced by $(s-a)$. Therefore,

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$$

■

Example 3.2.5. Find the Laplace transform of $f(t) = e^{-3t}\sin(4t)$.

Solution. We first identify the base function, which is $f_1(t) = \sin(4t)$. Its transform is $F_1(s) = \mathcal{L}\{\sin(4t)\} = \frac{4}{s^2+16}$. Now, we apply the First Translation Theorem with $a = -3$.

$$\mathcal{L}\{e^{-3t}\sin(4t)\} = F_1(s - (-3)) = F_1(s+3)$$

We replace every s in $F_1(s)$ with $(s+3)$:

$$F(s) = \frac{4}{(s+3)^2 + 16} = \frac{4}{s^2 + 6s + 9 + 16} = \frac{4}{s^2 + 6s + 25}$$

■

3.2.3 Change of Scale Property

This property describes how the transform behaves when the time variable is scaled by a constant factor.

Theorem 3.2.6 (Change of Scale). *If $\mathcal{L}\{f(t)\} = F(s)$, then for any positive constant a :*

$$\mathcal{L}\{f(at)\} = \frac{1}{a}F\left(\frac{s}{a}\right)$$

Proof. Starting from the definition:

$$\mathcal{L}\{f(at)\} = \int_0^\infty e^{-st} f(at) dt$$

Let's make a substitution $\tau = at$. Then $t = \tau/a$ and $dt = d\tau/a$. The limits of integration remain unchanged.

$$\begin{aligned} \mathcal{L}\{f(at)\} &= \int_0^\infty e^{-s(\tau/a)} f(\tau) \frac{d\tau}{a} \\ &= \frac{1}{a} \int_0^\infty e^{-(s/a)\tau} f(\tau) d\tau \end{aligned}$$

This is the definition of the Laplace transform of $f(\tau)$ with s replaced by (s/a) , multiplied by the constant $1/a$.

$$\mathcal{L}\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right)$$

■

3.3 Transforms of Special Functions

3.3.1 Laplace Transform of Periodic Functions

For a function that repeats itself over a fixed period, we do not need to integrate over an infinite interval. A special formula allows us to find the transform by integrating over just one period.

Theorem 3.3.1 (Transform of a Periodic Function). *Let $f(t)$ be a periodic function with period $T > 0$, so that $f(t+T) = f(t)$. Then the Laplace transform of $f(t)$ is given by:*

$$\mathcal{L}\{f(t)\} = \frac{\int_0^T e^{-st} f(t) dt}{1 - e^{-sT}}$$

Proof. We begin by splitting the integral from the definition into a sum of integrals over each period:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt = \int_0^T e^{-st} f(t) dt + \int_T^{2T} e^{-st} f(t) dt + \dots$$

Let's examine a generic term in the sum, $\int_{nT}^{(n+1)T} e^{-st} f(t) dt$. We use the substitution $t = u + nT$, so $dt = du$. As t goes from nT to $(n+1)T$, u goes from 0 to T .

$$\int_{nT}^{(n+1)T} e^{-st} f(t) dt = \int_0^T e^{-s(u+nT)} f(u+nT) du$$

Since $f(u+nT) = f(u)$ due to periodicity, and $e^{-s(u+nT)} = e^{-su}e^{-snT}$, this becomes:

$$e^{-snT} \int_0^T e^{-su} f(u) du$$

The full transform is the sum over all $n = 0, 1, 2, \dots$:

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \sum_{n=0}^{\infty} \left(e^{-snT} \int_0^T e^{-su} f(u) du \right) \\ &= \left(\sum_{n=0}^{\infty} e^{-snT} \right) \left(\int_0^T e^{-st} f(t) dt \right) \end{aligned}$$

The sum is a geometric series $\sum_{n=0}^{\infty} (e^{-sT})^n$, which converges to $\frac{1}{1-e^{-sT}}$ for $s > 0$. Substituting this back gives the theorem's result. ■

3.3.2 The Dirac Delta Function

The Dirac delta function, denoted $\delta(t)$, is not a function in the traditional sense but a *generalized function* or *distribution*. It is used to model phenomena of an impulsive nature, such as a point force, a point charge, or a signal of infinitesimal duration and immense amplitude.

Definition 3.3.2 (Dirac Delta Function). The Dirac delta function $\delta(t - a)$ is defined by its effect under an integral, known as the **sifting property**. For any continuous function $f(t)$:

$$\int_{-\infty}^{\infty} f(t)\delta(t - a) dt = f(a)$$

The function $\delta(t - a)$ can be conceptualized as an infinitely high, infinitesimally narrow spike at $t = a$ with a total area of 1. That is,

$$\delta(t - a) = 0 \text{ for } t \neq a, \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t - a) dt = 1$$

Theorem 3.3.3 (Laplace Transform of the Dirac Delta Function). For $a \geq 0$, the Laplace transform of the shifted Dirac delta function is:

$$\mathcal{L}\{\delta(t - a)\} = e^{-as}$$

A special case is the transform of the delta function at the origin:

$$\mathcal{L}\{\delta(t)\} = 1$$

Proof. We use the definition of the Laplace transform and the sifting property of the delta function. The function being "sifted" is e^{-st} .

$$\mathcal{L}\{\delta(t - a)\} = \int_0^{\infty} e^{-st}\delta(t - a) dt$$

Since the delta function is zero everywhere except at $t = a$, it "samples" the function e^{-st} at the point $t = a$.

$$\mathcal{L}\{\delta(t - a)\} = e^{-sa}$$

For the case $a = 0$, we simply have $e^{-0 \cdot s} = 1$. ■

Remark 3.3.4. The Dirac delta function is crucial in systems analysis. The response of a linear time-invariant system to an impulse input $\delta(t)$ is called the **impulse response** of the system. The Laplace transform of the impulse response is the system's **transfer function**, a fundamental concept in control theory and signal processing.

8.9 Laplace Transformation

$$\mathcal{L}[f'(t)] = s\mathcal{L}[f(t)] - f(0)$$

or

$$\frac{2}{s^2 + 4} = s\mathcal{L}[f(t)], \quad \text{or} \quad \mathcal{L}[f(t)] = \frac{2}{s(s^2 + 4)}$$

Example 8.11 Find the Laplace transform of $f(t) = t^3$, using the formula given in Eq. (8.11).

Solution We have $f(t) = t^3$, $f'(t) = 3t^2$, $f''(t) = 6t$, $f'''(t) = 6$ and $f(0) = f'(0) = f''(0) = 0$. Hence,

$$\mathcal{L}[f'''(t)] = \mathcal{L}[6] = s^3 \mathcal{L}[f(t)] - s^2 f(0) - s f'(0) - f''(0)$$

or

$$\frac{6}{s} = s^3 \mathcal{L}[f(t)], \quad \text{or} \quad \mathcal{L}[f(t)] = \frac{6}{s^4}$$

Example 8.12 Find $\mathcal{L}^{-1}[s/(s^2 + 9)]$ using the formula given in Eq. (8.9).

Solution Let $F(s) = 1/(s^2 + 9)$. Then $f(t) = \mathcal{L}^{-1}[F(s)] = \mathcal{L}^{-1}\left[\frac{1}{s^2 + 9}\right] = \frac{1}{3} \sin 3t$ and $f(0) = 0$. Therefore, using Eq. (8.9), we obtain

$$\mathcal{L}^{-1}[sF(s)] = f'(t) = \frac{d}{dt} \left(\frac{1}{3} \sin 3t \right) = \cos 3t.$$

We are now in a position to outline the procedure for solving the initial value problem

$$ay'' + by' + cy = f(t), \quad a, b, c \text{ constants}$$

$$y(0) = y_0, y'(0) = y'_0.$$

Taking the Laplace transform on both sides and denoting $\mathcal{L}[y(t)] = Y(s)$, and $\mathcal{L}[f(t)] = F(s)$, we obtain

$$a\mathcal{L}[y''] + b\mathcal{L}[y'] + c\mathcal{L}[y] = \mathcal{L}[f(t)]$$

$$a[s^2Y(s) - sy(0) - y'(0)] + b[sY(s) - y(0)] + cY(s) = F(s) \tag{3.1}$$

$$(as^2 + bs + c)Y(s) + F(s) + a(sy(0) + y'(0)) + by(0) \tag{3.2}$$

$$Y(s) = \frac{F(s) + a(sy(0) + y'(0)) + by(0)}{as^2 + bs + c} \tag{3.3}$$

which is an algebraic equation for $Y(s)$. Eq. (8.16) is called the subsidiary equation. Taking the inverse Laplace transform, we obtain

$$y(t) = \mathcal{L}^{-1}[Y(s)] = \mathcal{L}^{-1} \left[\frac{F(s) + a(sy(0) + y'(0)) + by(0)}{as^2 + bs + c} \right] \quad (3.4)$$

We need more results to obtain the inverse Laplace transform, given in Eq. (8.18), in the general case. However, in this section, we shall solve some simple problems.

Example 8.13 Solve the initial value problem

$$y'' + 4y = 0, \quad y(0) = 1, \quad y'(0) = 6.$$

Solution Taking Laplace transform of the given differential equation, we obtain

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In the limit, the two integrals on the right hand side of Eq. (8.8) combine to give

$$\int_0^\infty e^{-st} f'(t) dt = s \mathcal{L}[f(t)].$$

In the limit, the first term on the right hand side of Eq. (8.8) is $e^{-sT} f(T) - f(0^+)$ and the third term gives $-e^{-sT} f(T^-)$. Since, $f(t)$ is continuous on $[0, \infty)$, we have $f(0^+) = f(0)$ and $f(T^+) = f(T^-)$. Hence, the first and third terms give $-f(0)$. Substituting in Eq. (8.8), we obtain

$$\mathcal{L}[f'(t)] = s \mathcal{L}[f(t)] - f(0).$$

Remark 2 Let $f(t)$ satisfy the conditions given in Theorem 8.3 and $f(0) = 0$. Then

$$\mathcal{L}[f'(t)] = sF(s), \quad \text{or} \quad \mathcal{L}^{-1}[sF(s)] = f'(t) = \frac{d}{dt} \mathcal{L}^{-1}[F(s)]$$

where $\mathcal{L}[f(t)] = F(s)$. This result is useful in finding the inverse Laplace transform when the given function $\Phi(s)$ can be written as $sF(s)$. The result given in Eq. (8.7) can be extended to find the Laplace transform of higher order derivatives. We have

$$\mathcal{L}[f''(t)] = s \mathcal{L}[f'(t)] - f'(0) = s[s \mathcal{L}[f(t)] - f(0)] - f'(0) = s^2 \mathcal{L}[f(t)] - sf(0) - f'(0) \quad (3.5)$$

$$\begin{aligned} \mathcal{L}[f'''(t)] &= s \mathcal{L}[f''(t)] - f''(0) = s[s^2 \mathcal{L}[f(t)] - sf(0) - f'(0)] - f''(0) \\ &= s^3 \mathcal{L}[f(t)] - s^2 f(0) - sf'(0) - f''(0). \end{aligned} \quad (3.6)$$

By induction, we can prove the following result.

Theorem 8.4 Let $f(t), f'(t), \dots, f^{(n-1)}(t)$ be continuous on $[0, \infty)$ and be of exponential order. Let $f^{(n)}(t)$ be at least piecewise continuous on $[0, \infty)$. Then

$$\mathcal{L}[f^{(n)}(t)] = s^n \mathcal{L}[f(t)] - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0). \quad (3.7)$$

The above results, (Eqs. (8.9) to (8.12)) allow us to compute the Laplace transform of many functions which satisfy the given conditions.

Remark 3 Let $f(t)$ satisfy the conditions given in Theorem 8.3 and $f(0) = 0$ and $f'(0) = 0$. Then, from Eq. (8.10), we have

$$\mathcal{L}[f''(t)] = s^2 \mathcal{L}[f(t)], \quad \text{or} \quad \mathcal{L}^{-1}[s^2 F(s)] = f''(t) = \frac{d^2}{dt^2} \mathcal{L}^{-1}[F(s)]. \quad (3.8)$$

If $f(0) = f'(0) = \dots = f^{(n-1)}(0) = 0$, then from Eq. (8.12), we have

$$\mathcal{L}[f^{(n)}(t)] = s^n \mathcal{L}[f(t)], \quad \text{or} \quad \mathcal{L}^{-1}[s^n F(s)] = f^{(n)}(t). \quad (3.9)$$

Example 8.10 Find the Laplace transform of $f(t) = \sin^2 t$ using the differentiation formula given in Eq. (8.7).

Solution We have $f(t) = \sin^2 t$ and $f(0) = 0$. Differentiating, we obtain $f'(t) = 2 \sin t \cos t = \sin 2t$. Using the formula given in Eq. (8.7), we obtain

Laplace Transformation 8.7

37. $\frac{3}{s^2+2s+5}$

38. $\frac{13}{s^2-9}$

39. $\frac{2}{(s-1)(s-3)}$

40. $\frac{s}{(s+1)(s-3)}$

$$\mathcal{L}[t^a] = \frac{\Gamma(1+a)}{s^{a+1}}, \quad a > -1.$$

that Laplace transform of the function $f(t) = t^{1/2}$ given that $\Gamma(1/2) = \sqrt{\pi}$. $F(s) = \mathcal{L}[f(t)]$, show that $\mathcal{L}[af(t)] = a \mathcal{L}[f(t)]$.

44. Show that $\int_0^{n\pi} e^{-st} f(t) dt = \int_0^\pi e^{-st} f(t) dt$, n is an integer, when $f(t)$ is a periodic function of period T .

45. Find $\mathcal{L}[e^{at} \cos bt]$. Hence, show that

$$\mathcal{L}[e^{at} \cos bt] = \frac{s-a}{(s-a)^2 + b^2}, \quad \mathcal{L}[e^{at} \sin bt] = \frac{b}{(s-a)^2 + b^2}.$$

8.3 Laplace Transform Solution of Initial Value Problems

Laplace transformation is one of the important tools of solving the initial value problems. We first prove the following results on which the method of solution depends.

8.3.1 Laplace Transform of Derivatives

Theorem 8.3 Let $f(t), t \geq 0$ be a continuous function and be of exponential order on $[0, \infty)$. Let $f'(t)$ be also of exponential order and at least piecewise continuous on $[0, \infty)$. Then

$$\mathcal{L}[f'(t)] = s \mathcal{L}[f(t)] - f(0). \quad (3.10)$$

Proof First, consider the case when $f'(t)$ is continuous on $[0, \infty)$. Then

$$\begin{aligned} \mathcal{L}[f'(t)] &= \int_0^\infty e^{-st} f'(t) dt = [e^{-st} f(t)]_0^\infty + s \int_0^\infty e^{-st} f(t) dt \\ &= -f(0) + s \mathcal{L}[f(t)], \quad s > \alpha \quad (\text{see Theorem 8.2}). \end{aligned}$$

Hence, the result.

Suppose now, that $f'(t)$ is piecewise continuous. Assume that at $t = T$, $f'(t)$ has a finite jump. Then, we write

$$\begin{aligned} \mathcal{L}[f'(t)] &= \int_0^\infty e^{-st} f'(t) dt = \lim_{\epsilon \rightarrow 0} \int_0^{T-\epsilon} e^{-st} f'(t) dt + \int_{T+\epsilon}^\infty e^{-st} f'(t) dt \\ &= \lim_{\epsilon \rightarrow 0} \left([e^{-st} f(t)]_0^{T-\epsilon} + s \int_0^{T-\epsilon} e^{-st} f(t) dt \right) + \lim_{\epsilon \rightarrow 0} \left([e^{-st} f(t)]_{T+\epsilon}^\infty + s \int_{T+\epsilon}^\infty e^{-st} f(t) dt \right) \end{aligned} \quad (3.11)$$

Laplace Transformation 8.5

It is important to note that the above theorem gives only the sufficient conditions for the existence of the Laplace transform. That is, a function may have Laplace transform even if it violates the existence conditions. For example, consider the function $f(t) = t^{-1/2}$. This function is not continuous on any interval $[0, T)$ because it has limit as $t \rightarrow 0$ from the right. However, $\int_0^T e^{-st} t^{-1/2} dt$ exists for all $T > 0$. The Laplace transform of $t^{-1/2}$ can be computed as

$$\begin{aligned} \mathcal{L}[t^{-1/2}] &= \int_0^\infty e^{-st} t^{-1/2} dt = 2 \int_0^\infty e^{-u^2} du \quad (\text{setting } t = u^2) \\ &= \frac{2}{\sqrt{s}} \int_0^\infty e^{-x^2} dx \quad (\text{by setting } z = u\sqrt{s}) \\ &= \sqrt{\pi/s}, \quad s > 0. \end{aligned}$$

Remark 1 Let $f(t)$ satisfy the conditions of the existence theorem and $\mathcal{L}[f(t)] = F(s)$. Then

1. $\lim_{s \rightarrow \infty} F(s) = 0$.

2. $\lim_{s \rightarrow \infty} [sF(s)]$ is bounded.

These two results indicate that not all functions of s are Laplace transforms of some function $f(t)$.

Example 8.6 Find the Laplace transform of the piecewise continuous function

$$f(t) = \begin{cases} 0, & 0 \leq t < 2 \\ k, & t \geq 2, \text{ k constant.} \end{cases}$$

Solution We have

$$\mathcal{L}[f(t)] = \int_0^\infty e^{-st} f(t) dt = \int_2^\infty k e^{-st} dt = k \left[-\frac{e^{-st}}{s} \right]_2^\infty = \frac{k e^{-2s}}{s}, \quad s > 0.$$

Example 8.7 Find the Laplace transform of $\cos^2(at)$.

Solution We have $\cos^2 at = (1 + \cos 2at)/2$. Therefore,

$$\mathcal{L}[\cos^2 at] = \mathcal{L} \left[\frac{1}{2} + \frac{1}{2} \cos 2at \right] = \frac{1}{2s} + \frac{1}{2} \frac{s}{s^2 + (2a)^2} = \frac{1}{2s} + \frac{s}{2(s^2 + 4a^2)} = \frac{s^2 + 2a^2}{s(s^2 + 4a^2)}.$$

Example 8.8 Find $\mathcal{L}^{-1} \left[\frac{2s+5}{s^2+2s+25} \right]$. **Solution** We write $\frac{2s+5}{s^2+2s+25} = \frac{2s+5}{(s+1)^2+24}$. Then we have

$$\begin{aligned} \mathcal{L}^{-1} \left[\frac{2s+5}{(s+1)^2+24} \right] &= \mathcal{L}^{-1} \left[\frac{2(s+1)+3}{(s+1)^2+24} \right] \\ &= 2 \mathcal{L}^{-1} \left[\frac{s+1}{(s+1)^2+24} \right] + \frac{3}{\sqrt{24}} \mathcal{L}^{-1} \left[\frac{\sqrt{24}}{(s+1)^2+24} \right] \\ &= 2e^{-t} \cos(\sqrt{24}t) + \frac{3}{\sqrt{24}} e^{-t} \sin(\sqrt{24}t). \end{aligned}$$

Example 8.9 Find $\mathcal{L}^{-1} \left[\frac{5s^2+3s-16}{(s-1)(s-2)(s+3)} \right]$. **Solution** We use partial fractions to write the given function $F(s)$ as the sum of three factors.

Definition 3.3.5.

For example, since

$$|t| \leq e^{2t}, |e^{2t}| \leq e^{2t}, |\cos t| \leq 1, t \geq 0$$

the functions $f_1(t) = t$, $f_2(t) = e^{2t}$ and $f_3(t) = \cos t$ are all of exponential order. For a function of the form $f(t) = e^{ct^2}$, $c > 0$, it is not possible to determine constants α and M such that $e^{ct^2} \leq M e^{\alpha t}$. Therefore, e^{ct^2} is not a function of exponential order.

Theorem 8.2 (Sufficient conditions for existence of Laplace transform) If $f(t)$ is a piecewise continuous function on the interval $[0, \infty)$ and is of exponential order α for $t \geq 0$, then $\mathcal{L}[f(t)]$ exists for $s > \alpha$.

Proof If $f(t)$ is piecewise continuous on $[0, T]$ for $T > 0$, then $e^{-st} f(t)$ is also piecewise continuous on $[0, T]$. Hence, $\int_0^T e^{-st} f(t) dt$ exists. Therefore, the existence of the Laplace transform of $f(t)$ depends on whether this integral converges or has a finite limit as $T \rightarrow \infty$. To prove the convergence, we use the following results from the theory of improper integrals.

If for all t ($t \geq t_0$) the inequality $0 \leq \bar{f}(t) \leq g(t)$ is satisfied and if $\int_{t_0}^\infty g(t) dt$ converges, then $\int_{t_0}^\infty \bar{f}(t) dt$ also converges. If $\int_{t_0}^\infty |\bar{f}(t)| dt$ converges, then the integral $\int_{t_0}^\infty \bar{f}(t) dt$ also converges. In this case, the later integral is called an absolutely convergent integral. To use these results, we can determine a function $g(t)$ such that $\int_0^\infty g(t) dt$ converges and

$$|e^{-st} f(t)| \leq g(t), \quad t \geq 0.$$

Now, choose some numbers a and M such that

$$|f(t)| \leq M e^{at}.$$

Then, from Eq. (8.6), we obtain

$$|e^{-st} f(t)| \leq M e^{(a-s)t} = g(t).$$

Now,

$$\int_0^\infty g(t) dt = \int_0^\infty M e^{(a-s)t} dt$$

converges if $s > a$. Therefore, if we choose $g(t) = M e^{(a-s)t}$, $s > a$, the integral $\int_0^\infty e^{-st} f(t) dt$ is absolutely convergent and hence convergent. Hence, $\mathcal{L}[f(t)]$ exists. Alternatively, we have

$$\begin{aligned} |\mathcal{L}[f(t)]| &= \left| \int_0^\infty e^{-st} f(t) dt \right| \leq \int_0^\infty |e^{-st} f(t)| dt \leq M \int_0^\infty e^{(a-s)t} dt \\ &= M \left[\frac{e^{(a-s)t}}{a-s} \right]_0^\infty = \frac{M}{s-a}, \quad s > a. \end{aligned}$$

proving the existence of the Laplace transform.

3.4 The Inverse Laplace Transform

The process of finding the original function $f(t)$ from its transform $F(s)$ is called the inverse Laplace transform.

Definition 3.4.1 (Inverse Laplace Transform). If $F(s)$ is the Laplace transform of a function $f(t)$, i.e., $\mathcal{L}\{f(t)\} = F(s)$, then we say that $f(t)$ is the **inverse Laplace transform** of $F(s)$, and we write:

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

For our purposes, we find the inverse transform by recognizing the form of $F(s)$ in a table of known Laplace transform pairs and using its properties. The uniqueness of the inverse transform (for continuous functions) is guaranteed by Lerch's Theorem.

Theorem 3.4.2 (Linearity of the Inverse Transform). *The inverse Laplace transform is a linear operator. For any two transforms $F_1(s)$ and $F_2(s)$ and constants c_1, c_2 :*

$$\mathcal{L}^{-1}\{c_1 F_1(s) + c_2 F_2(s)\} = c_1 \mathcal{L}^{-1}\{F_1(s)\} + c_2 \mathcal{L}^{-1}\{F_2(s)\} = c_1 f_1(t) + c_2 f_2(t)$$

Proof. The proof follows directly from the linearity of the forward transform. Let $f_1(t) = \mathcal{L}^{-1}\{F_1(s)\}$ and $f_2(t) = \mathcal{L}^{-1}\{F_2(s)\}$. Then $\mathcal{L}\{c_1 f_1(t) + c_2 f_2(t)\} = c_1 \mathcal{L}\{f_1(t)\} + c_2 \mathcal{L}\{f_2(t)\} = c_1 F_1(s) + c_2 F_2(s)$. Taking the inverse transform of both sides yields the theorem. ■

Example 3.4.3 (Finding an Inverse Transform). Find the inverse Laplace transform of $F(s) = \frac{5}{s-3} - \frac{2s}{s^2+4}$.

Solution. Using the linearity property, we can find the inverse of each term separately.

$$\begin{aligned} f(t) &= \mathcal{L}^{-1}\left\{\frac{5}{s-3} - \frac{2s}{s^2+4}\right\} \\ &= 5\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\} - 2\mathcal{L}^{-1}\left\{\frac{s}{s^2+2^2}\right\} \end{aligned}$$

From our table of elementary transforms, we recognize these forms:

$$\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at} \quad \text{and} \quad \mathcal{L}^{-1}\left\{\frac{s}{s^2+a^2}\right\} = \cos(at)$$

Therefore,

$$f(t) = 5e^{3t} - 2\cos(2t)$$

Remark 3.4.4 (Method of Partial Fractions). Often, $F(s)$ is a rational function that does not directly appear in our table. In such cases, the method of **partial fraction decomposition** is used to break down the complex fraction into a sum of simpler fractions whose inverse transforms are known. This is the most common technique for finding inverse Laplace transforms in practice.

Example 3.4.5 (Inverse Transform via Partial Fractions). Find the inverse Laplace transform of $F(s) = \frac{3s+1}{s^2-s-6}$.

Solution. The denominator factors as $(s-3)(s+2)$. We decompose $F(s)$ into partial fractions:

$$\frac{3s+1}{(s-3)(s+2)} = \frac{A}{s-3} + \frac{B}{s+2}$$

To find A , we use the cover-up method: multiply by $(s-3)$ and set $s=3$:

$$A = \left. \frac{3s+1}{s+2} \right|_{s=3} = \frac{3(3)+1}{3+2} = \frac{10}{5} = 2$$

To find B , multiply by $(s+2)$ and set $s=-2$:

$$B = \left. \frac{3s+1}{s-3} \right|_{s=-2} = \frac{3(-2)+1}{-2-3} = \frac{-5}{-5} = 1$$

So, $F(s) = \frac{2}{s-3} + \frac{1}{s+2}$. Now we can easily find the inverse transform:

$$\begin{aligned} f(t) &= \mathcal{L}^{-1}\left\{\frac{2}{s-3} + \frac{1}{s+2}\right\} \\ &= 2\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s-(-2)}\right\} \\ &= 2e^{3t} + e^{-2t} \end{aligned}$$

Example 3.4.6 (Irreducible Quadratic Factor and s-Shifting). Find the inverse Laplace transform of $F(s) = \frac{s+5}{s^2+2s+5}$.

Solution. First, we inspect the denominator $s^2 + 2s + 5$. The discriminant is $b^2 - 4ac = 2^2 - 4(1)(5) = 4 - 20 = -16 < 0$. Since the discriminant is negative, the denominator has complex roots and is an irreducible quadratic over the reals. This signals that we should complete the square and use the First Translation (s-Shifting) Theorem in reverse.

Step 1: Complete the square in the denominator.

$$s^2 + 2s + 5 = (s^2 + 2s + 1) + 4 = (s + 1)^2 + 4 = (s + 1)^2 + 2^2$$

This gives

$$\begin{aligned} F(s) &= \frac{s + 5}{(s + 1)^2 + 2^2} \\ &= \frac{(s + 1) + 4}{(s + 1)^2 + 2^2} \\ &= \frac{s + 1}{(s + 1)^2 + 2^2} + \frac{4}{(s + 1)^2 + 2^2} \end{aligned}$$

We now find the inverse of each term.

- For the first term, we recognize the pattern for a shifted cosine:

$$\mathcal{L}^{-1} \left\{ \frac{s + 1}{(s + 1)^2 + 2^2} \right\} \quad \text{matches the form } \mathcal{L}^{-1} \left\{ \frac{s - a}{(s - a)^2 + b^2} \right\} = e^{at} \cos(bt)$$

Here, $a = -1$ and $b = 2$. So, the inverse is $e^{-t} \cos(2t)$.

- For the second term, we recognize the pattern for a shifted sine:

$$\mathcal{L}^{-1} \left\{ \frac{4}{(s + 1)^2 + 2^2} \right\} = 2 \cdot \mathcal{L}^{-1} \left\{ \frac{2}{(s + 1)^2 + 2^2} \right\}$$

This matches the form $2 \cdot \mathcal{L}^{-1} \left\{ \frac{b}{(s - a)^2 + b^2} \right\} = 2 \cdot e^{at} \sin(bt)$. With $a = -1$ and $b = 2$, the inverse is $2e^{-t} \sin(2t)$.

Combining the results, the final inverse transform is:

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = e^{-t} \cos(2t) + 2e^{-t} \sin(2t)$$

■

Example 3.4.7 (Repeated Linear Factor). Find the inverse Laplace transform of $F(s) = \frac{1}{s^2(s+1)}$.

Solution. The denominator has a repeated linear factor, s^2 . The partial fraction decomposition must take the form:

$$\frac{1}{s^2(s+1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1}$$

Multiplying both sides by the denominator $s^2(s+1)$ gives:

$$1 = As(s+1) + B(s+1) + Cs^2$$

We can find the constants using convenient values of s :

- Let $s = 0$: $1 = A(0) + B(0+1) + C(0) \implies B = 1$.
- Let $s = -1$: $1 = A(0) + B(0) + C(-1)^2 \implies C = 1$.
- To find A , we can expand and equate coefficients or substitute another value. Let's expand the terms with s^2 : $1 = As^2 + \dots + Cs^2$. The coefficient of s^2 on the left is 0, so $A + C = 0$. Since $C = 1$, we must have $A = -1$.

Thus, our transform is decomposed as:

$$F(s) = -\frac{1}{s} + \frac{1}{s^2} + \frac{1}{s+1}$$

Now we find the inverse transform of each simple term using linearity:

$$\begin{aligned} f(t) &= \mathcal{L}^{-1} \left\{ -\frac{1}{s} + \frac{1}{s^2} + \frac{1}{s+1} \right\} \\ &= -\mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} + \mathcal{L}^{-1} \left\{ \frac{1}{s^2} \right\} + \mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\} \end{aligned}$$

Using the standard table of transforms ($\mathcal{L}\{1\} = 1/s$, $\mathcal{L}\{t\} = 1/s^2$, $\mathcal{L}\{e^{-t}\} = 1/(s+1)$):

$$f(t) = -1 + t + e^{-t}$$

■

Example 3.4.8 (Distinct Quadratic Factors). Find the inverse Laplace transform of $F(s) = \frac{s}{(s^2+1)(s^2+4)}$.

Solution. This expression can be decomposed using partial fractions. Since the factors are quadratic, the numerators must be linear terms:

$$\frac{s}{(s^2+1)(s^2+4)} = \frac{As+B}{s^2+1} + \frac{Cs+D}{s^2+4}$$

A faster method for this specific form (only even powers of s) is to let $x = s^2$ and decompose the algebraic fraction first:

$$\frac{1}{(x+1)(x+4)} = \frac{P}{x+1} + \frac{Q}{x+4}$$

Using the cover-up method:

- $P = \frac{1}{x+4} \Big|_{x=-1} = \frac{1}{3}$
- $Q = \frac{1}{x+1} \Big|_{x=-4} = -\frac{1}{3}$

So, $\frac{1}{(x+1)(x+4)} = \frac{1/3}{x+1} - \frac{1/3}{x+4}$. Now substitute $x = s^2$ back and multiply by the original numerator s :

$$F(s) = s \left[\frac{1/3}{s^2+1} - \frac{1/3}{s^2+4} \right] = \frac{1}{3} \frac{s}{s^2+1} - \frac{1}{3} \frac{s}{s^2+4}$$

This form is now easily invertible.

$$\begin{aligned} f(t) &= \mathcal{L}^{-1} \left\{ \frac{1}{3} \frac{s}{s^2+1^2} - \frac{1}{3} \frac{s}{s^2+2^2} \right\} \\ &= \frac{1}{3} \mathcal{L}^{-1} \left\{ \frac{s}{s^2+1^2} \right\} - \frac{1}{3} \mathcal{L}^{-1} \left\{ \frac{s}{s^2+2^2} \right\} \end{aligned}$$

Recognizing the form for $\cos(at)$:

$$f(t) = \frac{1}{3} \cos(t) - \frac{1}{3} \cos(2t)$$

■

3.4.1 Laplace Transform of Derivatives

The single most important property for solving differential equations is the ability of the Laplace transform to convert differentiation in the time domain into multiplication by s in the frequency domain.

Theorem 3.4.9 (Transform of the First Derivative). *Let $f(t)$ be continuous and of exponential order, and let $f'(t)$ be piecewise continuous for $t \geq 0$. Then the Laplace transform of the derivative $f'(t)$ is given by:*

$$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0) = sF(s) - f(0)$$

Proof. We use the definition of the Laplace transform and apply integration by parts.

$$\mathcal{L}\{f'(t)\} = \int_0^\infty e^{-st} f'(t) dt$$

Let $u = e^{-st}$ and $dv = f'(t)dt$. Then $du = -se^{-st}dt$ and $v = f(t)$. Applying the integration by parts formula, $\int u dv = uv - \int v du$:

$$\begin{aligned} \mathcal{L}\{f'(t)\} &= [e^{-st} f(t)]_0^\infty - \int_0^\infty f(t)(-se^{-st} dt) \\ &= \left(\lim_{b \rightarrow \infty} e^{-sb} f(b) - e^0 f(0) \right) + s \int_0^\infty e^{-st} f(t) dt \end{aligned}$$

Since $f(t)$ is of exponential order, for a sufficiently large s , the limit term $\lim_{b \rightarrow \infty} e^{-sb} f(b)$ goes to zero. The second term is $-f(0)$, and the integral is by definition $s\mathcal{L}\{f(t)\}$. Therefore,

$$\mathcal{L}\{f'(t)\} = -f(0) + sF(s) = sF(s) - f(0)$$

■

This theorem can be extended to higher-order derivatives.

Theorem 3.4.10 (Transform of the Second Derivative). *If $f(t)$ and $f'(t)$ are continuous and $f''(t)$ is piecewise continuous, all of exponential order, then:*

$$\mathcal{L}\{f''(t)\} = s^2 F(s) - sf(0) - f'(0)$$

Proof. We apply the first derivative theorem twice. Let $g(t) = f'(t)$. Then $g'(t) = f''(t)$.

$$\begin{aligned}\mathcal{L}\{f''(t)\} &= \mathcal{L}\{g'(t)\} \\ &= s\mathcal{L}\{g(t)\} - g(0) \\ &= s\mathcal{L}\{f'(t)\} - f'(0) \\ &= s(sF(s) - f(0)) - f'(0) \\ &= s^2 F(s) - sf(0) - f'(0)\end{aligned}$$

■

The general formula for the n -th derivative is $\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - \dots - f^{(n-1)}(0)$.

3.4.2 Laplace Transform of Integrals

Conversely, integration in the time domain corresponds to division by s in the frequency domain.

Theorem 3.4.11 (Transform of an Integral). *If $f(t)$ is piecewise continuous and of exponential order, then:*

$$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{1}{s}\mathcal{L}\{f(t)\} = \frac{F(s)}{s}$$

Proof. Let the integral be denoted by $g(t) = \int_0^t f(\tau) d\tau$. By the Fundamental Theorem of Calculus, we have $g'(t) = f(t)$. Also, it is clear from the definition that $g(0) = \int_0^0 f(\tau) d\tau = 0$. Now we apply the transform of the first derivative theorem to $g(t)$:

$$\mathcal{L}\{g'(t)\} = s\mathcal{L}\{g(t)\} - g(0)$$

Substituting our known values:

$$\begin{aligned}\mathcal{L}\{f(t)\} &= s\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} - 0 \\ F(s) &= s\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\}\end{aligned}$$

Rearranging the equation for the desired term gives the result:

$$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}$$

■

We are now in a position to outline the procedure for solving the initial value problem using Laplace transforms:

$$\begin{aligned}ay'' + by' + cy &= f(t), \quad a, b, c \text{ constants} \\ y(0) &= y_0, y'(0) = y'_0.\end{aligned}$$

Taking the Laplace transform on both sides and denoting

$$\begin{aligned}\mathcal{L}[y(t)] &= Y(s), \quad \text{and} \quad \mathcal{L}[f(t)] = F(s), \quad \text{we obtain} \\ a\mathcal{L}[y''] + b\mathcal{L}[y'] + c\mathcal{L}[y] &= \mathcal{L}[f(t)] \\ \text{or } a[s^2 Y(s) - sy(0) - y'(0)] + b[sY(s) - y(0)] + cY(s) &= F(s) \\ \text{or } (as^2 + bs + c)Y(s) &= F(s) + a(sy(0) + y'(0)) + by(0) \\ \text{or } Y(s) &= \frac{F(s) + a(sy(0) + y'(0)) + by(0)}{as^2 + bs + c}\end{aligned}$$

which is an algebraic equation for $Y(s)$. Taking the inverse Laplace transform, we obtain

$$y(t) = \mathcal{L}^{-1}[Y(s)] = \mathcal{L}^{-1}\left[\frac{F(s) + a(sy(0) + y'(0)) + by(0)}{as^2 + bs + c}\right]$$

Example 3.4.12. Solve the initial value problem

$$y'' + 4y = 0, \quad y(0) = 1, \quad y'(0) = 6.$$

Solution. Taking Laplace transform of the given differential equation, we obtain $[s^2Y(s) - sy(0) - y'(0)] + 4Y(s) = 0$, where $\mathcal{L}[y(t)] = Y(s)$. Substituting the values of $y(0)$ and $y'(0)$, we obtain

$$\begin{aligned}(s^2 + 4)Y(s) &= s + 6, \\ \text{or } Y(s) &= \frac{s + 6}{s^2 + 4} \\ &= \frac{s}{s^2 + 4} + \frac{6}{s^2 + 4}.\end{aligned}$$

$$\text{Therefore, } y(t) = \mathcal{L}^{-1}[Y(s)] = \mathcal{L}^{-1}\left[\frac{s}{s^2 + 4}\right] + \mathcal{L}^{-1}\left[\frac{6}{s^2 + 4}\right] = \cos 2t + 3 \sin 2t.$$

■

Example 3.4.13. Solve the initial value problem

$$y'' + 2y' - 3y = 3, \quad y(0) = 4, \quad y'(0) = -7.$$

Solution. Taking Laplace transform of the differential equation, we obtain

$$\begin{aligned}[s^2Y(s) - sy(0) - y'(0)] + 2[sY(s) - y(0)] - 3Y(s) &= \mathcal{L}[3] \\ \text{or } (s^2 + 2s - 3)Y(s) &= \frac{3}{s} + 4s - 7 + 8 = \frac{3}{s} + 4s + 1 \\ \text{where } \mathcal{L}[y(t)] &= Y(s). \\ \text{Therefore, } Y(s) &= \frac{3}{s(s-1)(s+3)} + \frac{4s+1}{(s-1)(s+3)}.\end{aligned}$$

Using the partial fractions, we obtain

$$\begin{aligned}Y(s) &= \left[-\frac{1}{s} + \frac{3}{4} \frac{1}{s-1} - \frac{1}{4} \frac{1}{s+3}\right] + \left[\frac{5}{4} \frac{1}{s-1} + \frac{11}{4} \frac{1}{s+3}\right] \\ &= -\frac{1}{s} + \frac{2}{s-1} + \frac{3}{s+3}.\end{aligned}$$

Taking the inverse Laplace transform, we get

$$y(t) = \mathcal{L}^{-1}[Y(s)] = \mathcal{L}^{-1}\left(-\frac{1}{s}\right) + \mathcal{L}^{-1}\left(\frac{2}{s-1}\right) + \mathcal{L}^{-1}\left(\frac{3}{s+3}\right) = -1 + 2e^t + 3e^{-3t}.$$

■

Example 3.4.14. Solve the initial value problem $y'' + 4y' + 4y = e^{-2t}$ with $y(0) = 0$ and $y'(0) = 1$.

Solution. Let $Y(s) = \mathcal{L}\{y(t)\}$. Taking the Laplace transform of the entire equation gives:

$$\mathcal{L}\{y''\} + 4\mathcal{L}\{y'\} + 4\mathcal{L}\{y\} = \mathcal{L}\{e^{-2t}\}$$

Applying the derivative rules and substituting the initial conditions yields:

$$\begin{aligned}[s^2Y(s) - sy(0) - y'(0)] + 4[sY(s) - y(0)] + 4Y(s) &= \frac{1}{s+2} \\ [s^2Y(s) - 1] + 4sY(s) + 4Y(s) &= \frac{1}{s+2}\end{aligned}$$

Solving for $Y(s)$ algebraically:

$$\begin{aligned}Y(s)(s^2 + 4s + 4) &= 1 + \frac{1}{s+2} = \frac{s+3}{s+2} \\ Y(s) &= \frac{s+3}{(s+2)(s+2)^2} = \frac{s+3}{(s+2)^3}\end{aligned}$$

To find the inverse transform, we rewrite the expression as:

$$Y(s) = \frac{(s+2)+1}{(s+2)^3} = \frac{1}{(s+2)^2} + \frac{1}{(s+2)^3}$$

The inverse transform of each term is found using the s-Shifting Theorem, recalling that $\mathcal{L}^{-1}\{1/s^2\} = t$ and $\mathcal{L}^{-1}\{2/s^3\} = t^2$:

$$y(t) = e^{-2t}t + e^{-2t}\frac{t^2}{2} = te^{-2t} + \frac{1}{2}t^2e^{-2t}$$

■

Example 3.4.15. Solve the initial value problem

$$y'' - 5y' + 4y = e^{2t}, \quad y(0) = 19/12, y'(0) = 8/3.$$

Theorem 3.4.16. Let $f(t)$ be piecewise continuous on $[0, \infty)$ and be of exponential order. Further, let $\lim_{t \rightarrow 0^+} [f(t)/t]$ exist. Then

$$\mathcal{L}\left[\frac{f(t)}{t}\right] = \int_s^\infty F(s)ds, \quad s > \alpha$$

where $\mathcal{L}[f(t)] = F(s)$.

Proof. From the definition, we have

$$F(s) = \int_0^\infty e^{-st} f(t) dt.$$

Integrating from s to ∞ , we obtain

$$\int_s^\infty F(s)ds = \int_s^\infty \left[\int_0^\infty e^{-st} f(t) dt \right] ds.$$

Now, $f(t)$ is piecewise continuous, is of exponential order and limit of $f(t)/t$ exists as t approaches zero from the right. Under these assumptions, the order of integration can be reversed and we can write

$$\begin{aligned} \int_s^\infty F(s)ds &= \int_0^\infty \left[\int_s^\infty e^{-st} ds \right] f(t) dt \\ &= \int_0^\infty \left[\frac{e^{-st}}{-t} \right]_s^\infty f(t) dt \\ &= \int_0^\infty \left[e^{-st} \frac{f(t)}{t} \right] dt = \mathcal{L}\left[\frac{f(t)}{t}\right], \quad s > \alpha. \end{aligned}$$

■

Example 3.4.17. Find the Laplace transform of $(\sin \omega t)/t$.

Solution Using above Theorem, we obtain

$$\begin{aligned} \mathcal{L}\left[\frac{\sin \omega t}{t}\right] &= \int_s^\infty \mathcal{L}[\sin \omega t] ds \\ &= \int_s^\infty \frac{\omega}{s^2 + \omega^2} ds \\ &= \left[\tan^{-1} \left(\frac{s}{\omega} \right) \right]_s^\infty \\ &= \frac{\pi}{2} - \tan^{-1} \left(\frac{s}{\omega} \right) \\ &= \cot^{-1} \left(\frac{s}{\omega} \right). \end{aligned}$$

Definition 3.4.18 (Convolution Integral). The convolution of two functions $f(t)$ and $g(t)$, denoted by $(f * g)(t)$, is defined as:

$$(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau$$

Theorem 3.4.19 (Convolution Theorem). If $\mathcal{L}\{f(t)\} = F(s)$ and $\mathcal{L}\{g(t)\} = G(s)$, then the Laplace transform of their convolution is the product of their transforms:

$$\mathcal{L}\{(f * g)(t)\} = F(s)G(s)$$

In its inverse form, which is more commonly used for problem-solving:

$$\mathcal{L}^{-1}\{F(s)G(s)\} = (f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau$$

Proof. By the definition of the Laplace transform, we have:

$$\mathcal{L}\{(f * g)(t)\} = \int_0^\infty e^{-st} (f * g)(t) dt$$

Substituting the definition of the convolution integral, $(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau$:

$$\mathcal{L}\{(f * g)(t)\} = \int_0^\infty e^{-st} \left[\int_0^t f(\tau)g(t - \tau) d\tau \right] dt$$

This is a double integral over the region in the t - τ plane defined by $0 \leq \tau \leq t < \infty$. We can change the order of integration. The new limits will be $\tau \leq t < \infty$ for the inner integral and $0 \leq \tau < \infty$ for the outer integral.

$$\begin{aligned}\mathcal{L}\{(f * g)(t)\} &= \int_0^\infty \int_\tau^\infty e^{-st} f(\tau) g(t - \tau) dt d\tau \\ &= \int_0^\infty f(\tau) \left[\int_\tau^\infty e^{-st} g(t - \tau) dt \right] d\tau\end{aligned}$$

Let's evaluate the inner integral by making the substitution $u = t - \tau$. This implies $t = u + \tau$ and $dt = du$. The limits of integration for u become 0 to ∞ .

$$\begin{aligned}\int_\tau^\infty e^{-st} g(t - \tau) dt &= \int_0^\infty e^{-s(u+\tau)} g(u) du \\ &= \int_0^\infty e^{-su} e^{-s\tau} g(u) du \\ &= e^{-s\tau} \int_0^\infty e^{-su} g(u) du \\ &= e^{-s\tau} G(s)\end{aligned}$$

Substituting this result back into the outer integral:

$$\begin{aligned}\mathcal{L}\{(f * g)(t)\} &= \int_0^\infty f(\tau) [e^{-s\tau} G(s)] d\tau \\ &= G(s) \int_0^\infty e^{-s\tau} f(\tau) d\tau\end{aligned}$$

The remaining integral is, by definition, the Laplace transform of $f(t)$, which is $F(s)$. Therefore,

$$\mathcal{L}\{(f * g)(t)\} = G(s)F(s) = F(s)G(s)$$

This completes the proof. ■

Example 3.4.20. Find the inverse Laplace transform of $H(s) = \frac{1}{s(s^2+4)}$ using convolution.

Solution. We express $H(s)$ as the product of $F(s) = \frac{1}{s}$ and $G(s) = \frac{1}{s^2+4}$. The corresponding inverse transforms are $f(t) = 1$ and $g(t) = \frac{1}{2} \sin(2t)$. Applying the convolution theorem, the inverse transform $h(t)$ is given by the integral:

$$\begin{aligned}h(t) &= (g * f)(t) = \int_0^t g(\tau) f(t - \tau) d\tau = \int_0^t \frac{1}{2} \sin(2\tau) \cdot (1) d\tau \\ &= \frac{1}{2} \left[-\frac{\cos(2\tau)}{2} \right]_0^t = -\frac{1}{4} (\cos(2t) - \cos(0)) \\ &= \frac{1}{4} (1 - \cos(2t))\end{aligned}$$
■

Theorem 3.4.21. Using convolution, solve the initial value problem

$$y'' + 9y = \sin 3t, \quad y(0) = 0, \quad y'(0) = 0.$$

Solution. Taking Laplace transform of the differential equation, we obtain

$$s^2 Y(s) + 9Y(s) = \frac{3}{s^2 + 9}, \quad \text{or} \quad Y(s) = \frac{3}{(s^2 + 9)^2}.$$

Write $Y(s)$ as

$$Y(s) = \frac{1}{3} \left(\frac{3}{s^2 + 9} \right) \left(\frac{3}{s^2 + 9} \right) = \frac{1}{3} F(s)G(s)$$

where $F(s) = G(s) = 3/(s^2 + 9)$. We have $f(t) = g(t) = \sin 3t$. Using the convolution theorem, we obtain

$$\begin{aligned}
 y(t) &= \mathcal{L}^{-1} \left[\frac{1}{3} F(s) G(s) \right] = \frac{1}{3} \int_0^t \sin 3\tau \sin 3(t - \tau) d\tau \\
 &= \frac{1}{6} \int_0^t [\cos(3\tau - 3(t - \tau)) - \cos(3\tau + 3(t - \tau))] d\tau \\
 &= \frac{1}{6} \int_0^t [\cos(6\tau - 3t) - \cos 3t] d\tau \\
 &= \frac{1}{6} \left[\frac{\sin(6\tau - 3t)}{6} - \tau \cos 3t \right]_0^t \\
 &= \frac{1}{6} \left[\frac{\sin(6t - 3t)}{6} - t \cos 3t - \left(\frac{\sin(-3t)}{6} - 0 \right) \right] \\
 &= \frac{1}{6} \left[\frac{\sin 3t}{6} - t \cos 3t + \frac{\sin 3t}{6} \right] \\
 &= \frac{1}{6} \left[\frac{2 \sin 3t}{6} - t \cos 3t \right] \\
 &= \frac{1}{6} \left[\frac{1}{3} \sin 3t - t \cos 3t \right] \\
 &= \frac{1}{18} \sin 3t - \frac{1}{6} t \cos 3t.
 \end{aligned}$$

■

3.5 Applications to Partial Differential Equations

The Laplace transform can simplify linear partial differential equations by transforming one variable (typically time t), thereby reducing the PDE to an ODE in the remaining spatial variable(s). The key relations for a function $u(x, t)$ with transform $U(x, s) = \mathcal{L}\{u(x, t)\}$ are:

$$\mathcal{L} \left\{ \frac{\partial u}{\partial t} \right\} = sU(x, s) - u(x, 0) \quad \text{and} \quad \mathcal{L} \left\{ \frac{\partial^2 u}{\partial x^2} \right\} = \frac{d^2 U}{dx^2}$$

Example 3.5.1 (Application to the Heat Equation). Solve the heat conduction problem in a semi-infinite rod ($x \geq 0$) governed by:

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2} \quad \text{with } u(x, 0) = 0, \quad u(0, t) = u_0, \quad \lim_{x \rightarrow \infty} u(x, t) = 0$$

Solution. Applying the Laplace transform with respect to t to the PDE and its boundary conditions converts the problem into an ordinary differential equation for $U(x, s)$. The PDE becomes:

$$sU(x, s) - u(x, 0) = k \frac{d^2 U}{dx^2} \implies \frac{d^2 U}{dx^2} - \frac{s}{k} U = 0$$

The transformed boundary conditions are $U(0, s) = u_0/s$ and $\lim_{x \rightarrow \infty} U(x, s) = 0$. The general solution to this ODE is $U(x, s) = C_1 e^{\sqrt{s/k}x} + C_2 e^{-\sqrt{s/k}x}$. The boundary condition at infinity requires $C_1 = 0$ to ensure a bounded solution. The condition at $x = 0$ then gives $C_2 = u_0/s$. The solution in the s -domain is therefore:

$$U(x, s) = \frac{u_0}{s} e^{-x\sqrt{s/k}}$$

The inverse transform of this expression is non-trivial and can be found in standard tables:

$$u(x, t) = u_0 \operatorname{erfc} \left(\frac{x}{2\sqrt{kt}} \right)$$

where erfc is the complementary error function. The method successfully reduces the PDE to a solvable ODE. ■

Example 3.5.2. Find the inverse Laplace transform of $H(s) = \frac{s}{(s^2 + a^2)^2}$ using the Convolution Theorem.

Solution. We can express $H(s)$ as the product of $F(s) = \frac{s}{s^2 + a^2}$ and $G(s) = \frac{1}{s^2 + a^2}$. The inverse transforms of these functions are:

$$\begin{aligned}
 f(t) &= \mathcal{L}^{-1}\{F(s)\} = \cos(at) \\
 g(t) &= \mathcal{L}^{-1}\{G(s)\} = \frac{1}{a} \sin(at)
 \end{aligned}$$

By the Convolution Theorem, $h(t) = \mathcal{L}^{-1}\{H(s)\}$ is given by the convolution integral $(f * g)(t)$:

$$h(t) = \int_0^t f(\tau)g(t-\tau) d\tau = \int_0^t \cos(a\tau) \left(\frac{1}{a} \sin(a(t-\tau)) \right) d\tau$$

Using the product-to-sum trigonometric identity $\sin(A) \cos(B) = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$, where $A = a(t-\tau)$ and $B = a\tau$:

$$\begin{aligned} h(t) &= \frac{1}{a} \int_0^t \frac{1}{2} [\sin(at - a\tau + a\tau) + \sin(at - a\tau - a\tau)] d\tau \\ &= \frac{1}{2a} \int_0^t [\sin(at) + \sin(at - 2a\tau)] d\tau \end{aligned}$$

We integrate with respect to τ , treating t as a constant:

$$\begin{aligned} h(t) &= \frac{1}{2a} \left[\tau \sin(at) + \frac{\cos(at - 2a\tau)}{2a} \right]_0^t \\ &= \frac{1}{2a} \left[\left(t \sin(at) + \frac{\cos(-at)}{2a} \right) - \left(0 + \frac{\cos(at)}{2a} \right) \right] \\ &= \frac{1}{2a} \left[t \sin(at) + \frac{\cos(at)}{2a} - \frac{\cos(at)}{2a} \right] \\ &= \frac{t}{2a} \sin(at) \end{aligned}$$

This result is a standard entry in Laplace transform tables and is fundamental in the study of resonance. ■

Example 3.5.3. Solve the initial value problem $y'' + y = \cos(2t)$ with initial conditions $y(0) = 1$ and $y'(0) = 0$.

Solution. Let $Y(s) = \mathcal{L}\{y(t)\}$. Taking the Laplace transform of the equation and applying the initial conditions gives:

$$\begin{aligned} [s^2 Y(s) - sy(0) - y'(0)] + Y(s) &= \mathcal{L}\{\cos(2t)\} \\ s^2 Y(s) - s(1) - 0 + Y(s) &= \frac{s}{s^2 + 4} \end{aligned}$$

Now, we solve algebraically for $Y(s)$:

$$\begin{aligned} Y(s)(s^2 + 1) &= s + \frac{s}{s^2 + 4} \\ Y(s) &= \frac{s}{s^2 + 1} + \frac{s}{(s^2 + 1)(s^2 + 4)} \end{aligned}$$

We use partial fractions on the second term. As shown in a previous example, this decomposition yields:

$$\frac{s}{(s^2 + 1)(s^2 + 4)} = \frac{1}{3} \frac{s}{s^2 + 1} - \frac{1}{3} \frac{s}{s^2 + 4}$$

Substituting this back into the expression for $Y(s)$:

$$Y(s) = \frac{s}{s^2 + 1} + \left(\frac{1}{3} \frac{s}{s^2 + 1} - \frac{1}{3} \frac{s}{s^2 + 4} \right) = \frac{4}{3} \frac{s}{s^2 + 1} - \frac{1}{3} \frac{s}{s^2 + 4}$$

Finally, we find the inverse Laplace transform term-by-term:

$$\begin{aligned} y(t) &= \mathcal{L}^{-1} \left\{ \frac{4}{3} \frac{s}{s^2 + 1} - \frac{1}{3} \frac{s}{s^2 + 4} \right\} \\ &= \frac{4}{3} \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 1^2} \right\} - \frac{1}{3} \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 2^2} \right\} \\ &= \frac{4}{3} \cos(t) - \frac{1}{3} \cos(2t) \end{aligned}$$

The solution consists of the natural frequency oscillation $(\cos t)$ and an oscillation at the driving frequency $(\cos 2t)$. ■

Example 3.5.4 (Application to the Wave Equation). A semi-infinite elastic string ($x \geq 0$) is initially at rest and in its equilibrium position. At time $t = 0$, the end at $x = 0$ is forced to undergo sinusoidal motion given by $f(t) = \sin(\omega t)$. Find the resulting displacement $u(x, t)$.

Solution. The problem is described by the one-dimensional wave equation and its associated conditions:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Conditions: $u(x, 0) = 0$, $u_t(x, 0) = 0$, $u(0, t) = \sin(\omega t)$, and $\lim_{x \rightarrow \infty} u(x, t)$ is bounded.

Let $U(x, s) = \mathcal{L}\{u(x, t)\}$. We transform the PDE with respect to t :

$$s^2 U(x, s) - su(x, 0) - u_t(x, 0) = c^2 \frac{d^2 U}{dx^2}$$

Substituting the initial conditions simplifies this to an ODE in x :

$$s^2 U(x, s) = c^2 \frac{d^2 U}{dx^2} \implies \frac{d^2 U}{dx^2} - \left(\frac{s}{c}\right)^2 U = 0$$

The transformed boundary conditions are $U(0, s) = \mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2 + \omega^2}$ and $\lim_{x \rightarrow \infty} U(x, s) = 0$. The general solution to the ODE is $U(x, s) = C_1 e^{(s/c)x} + C_2 e^{-(s/c)x}$. The boundedness condition as $x \rightarrow \infty$ forces $C_1 = 0$. Applying the condition at $x = 0$ gives $C_2 = U(0, s) = \frac{\omega}{s^2 + \omega^2}$. The solution in the s -domain is:

$$U(x, s) = \frac{\omega}{s^2 + \omega^2} e^{-(x/c)s}$$

To find the inverse transform, we use the Second Translation (t-Shifting) Theorem, which states $\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t-a)H(t-a)$, where H is the Heaviside step function. Here, $a = x/c$ and $F(s) = \frac{\omega}{s^2 + \omega^2}$, so $f(t) = \sin(\omega t)$. Applying the theorem, the solution is:

$$u(x, t) = \sin\left(\omega\left(t - \frac{x}{c}\right)\right) H\left(t - \frac{x}{c}\right)$$

This represents a sinusoidal wave traveling to the right with speed c . The Heaviside function indicates that the displacement at a point x remains zero until time $t = x/c$, the time required for the wave to propagate from the origin to that point. ■

Finding Laplace Transforms

Find the Laplace transform, $F(s) = \mathcal{L}\{f(t)\}$, for each of the following functions.

1. $f(t) = 2t^3 - 4e^{-t} + 3\sin(5t)$
2. $f(t) = t^2 e^{3t}$
3. $f(t) = e^{-4t} \cosh(2t)$
4. $f(t) = \sin(t) \cos(t)$ (Hint: Use a trigonometric identity first.)
5. $f(t) = \frac{1-e^{-t}}{t}$ (Hint: Consider the transform of an integral.)
6. $f(t) = t^2 \cos(3t)$ (Hint: Use the property for multiplication by t^n .)
7. $f(t)$ is a periodic square wave with period $T = 2$ defined by:

$$f(t) = \begin{cases} 1, & 0 \leq t < 1 \\ -1, & 1 \leq t < 2 \end{cases}$$

8. $f(t) = 5\delta(t) + t^3\delta(t-2)$

Finding Inverse Laplace Transforms

Find the inverse Laplace transform, $f(t) = \mathcal{L}^{-1}\{F(s)\}$, for each of the following functions.

9. $F(s) = \frac{3s+7}{s^2-2s-3}$
10. $F(s) = \frac{s-1}{s^2(s+3)}$
11. $F(s) = \frac{3s-2}{s^2+2s+10}$
12. $F(s) = \frac{s}{(s^2+4)^2}$ (Use the Convolution Theorem.)
13. $F(s) = \frac{e^{-2s}}{s^2+9}$
14. $F(s) = \ln\left(1 + \frac{\omega^2}{s^2}\right)$ (Hint: Differentiate with respect to s first.)

Solve Ordinary Differential Equations

Use the Laplace transform to solve the following initial value problems.

15. $y'' - 6y' + 9y = t^2 e^{3t}$, with $y(0) = 2$ and $y'(0) = 6$.

16. $y'' + 4y = \sin(2t)$, with $y(0) = 1$ and $y'(0) = -1$.

17. $y'' - y' - 2y = 4t^2$, with $y(0) = 1$ and $y'(0) = 4$.

18. $y'(t) + \int_0^t y(\tau) d\tau = \cos(t)$, with $y(0) = 1$.

Solve Partial Differential Equations

Use the Laplace transform to solve the following boundary value problems.

19. The one-dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = 9 \frac{\partial^2 u}{\partial x^2}$ for $x > 0, t > 0$, subject to the conditions:

- $u(x, 0) = 0$ and $u_t(x, 0) = 0$
- $u(0, t) = t$ for $t \geq 0$
- $\lim_{x \rightarrow \infty} u(x, t) = 0$

20. The one-dimensional heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ for a semi-infinite rod ($x > 0, t > 0$), subject to the conditions:

- $u(x, 0) = 50$
- $u(0, t) = 0$ for $t > 0$
- $u(x, t)$ is bounded as $x \rightarrow \infty$.

Chapter 4

Tutorial - Integral Transforms

4.1 Applications to Differential and Integral Equations

Example 4.1.1 (Second-Order Homogeneous ODE). Solve the initial value problem $y'' - y' - 6y = 0$ with initial conditions $y(0) = 1$ and $y'(0) = -1$.

Solution. Let $Y(s) = \mathcal{L}\{y(t)\}$. Taking the Laplace transform of the equation and incorporating the initial conditions, we get:

$$\begin{aligned}[s^2Y(s) - sy(0) - y'(0)] - [sY(s) - y(0)] - 6Y(s) &= 0 \\ s^2Y(s) - s(1) - (-1) - sY(s) + 1 - 6Y(s) &= 0\end{aligned}$$

We group terms to solve for $Y(s)$:

$$\begin{aligned}Y(s)(s^2 - s - 6) &= s - 2 \\ Y(s) &= \frac{s - 2}{s^2 - s - 6} = \frac{s - 2}{(s - 3)(s + 2)}\end{aligned}$$

Using partial fraction decomposition:

$$\frac{s - 2}{(s - 3)(s + 2)} = \frac{A}{s - 3} + \frac{B}{s + 2}$$

The constants are found to be $A = \frac{3-2}{3+2} = \frac{1}{5}$ and $B = \frac{-2-2}{-2-3} = \frac{4}{5}$. So, $Y(s) = \frac{1}{5} \frac{1}{s-3} + \frac{4}{5} \frac{1}{s+2}$. Finding the inverse Laplace transform gives the solution:

$$y(t) = \frac{1}{5} \mathcal{L}^{-1} \left\{ \frac{1}{s-3} \right\} + \frac{4}{5} \mathcal{L}^{-1} \left\{ \frac{1}{s+2} \right\} = \frac{1}{5} e^{3t} + \frac{4}{5} e^{-2t}$$

■

Example 4.1.2 (Second-Order Non-Homogeneous ODE). Solve the initial value problem $y'' + 4y = 8t$ with $y(0) = 1$ and $y'(0) = 0$.

Solution. Transforming the equation gives:

$$\begin{aligned}[s^2Y(s) - sy(0) - y'(0)] + 4Y(s) &= \mathcal{L}\{8t\} \\ s^2Y(s) - s(1) - 0 + 4Y(s) &= \frac{8}{s^2}\end{aligned}$$

Solving for $Y(s)$:

$$Y(s)(s^2 + 4) = s + \frac{8}{s^2} \implies Y(s) = \frac{s}{s^2 + 4} + \frac{8}{s^2(s^2 + 4)}$$

We decompose the second term: $\frac{8}{s^2(s^2+4)} = \frac{2}{s^2} - \frac{2}{s^2+4}$.

$$Y(s) = \frac{s}{s^2 + 4} + \frac{2}{s^2} - \frac{2}{s^2 + 4}$$

Taking the inverse transform of each term:

$$\begin{aligned}y(t) &= \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 2^2} \right\} + 2\mathcal{L}^{-1} \left\{ \frac{1}{s^2} \right\} - \mathcal{L}^{-1} \left\{ \frac{2}{s^2 + 2^2} \right\} \\ &= \cos(2t) + 2t - \sin(2t)\end{aligned}$$

■

Example 4.1.3 (System of First-Order ODEs). Solve the system of differential equations for $t \geq 0$:

$$\begin{aligned}x'(t) &= -y(t) \\ y'(t) &= x(t)\end{aligned}$$

with initial conditions $x(0) = 1$ and $y(0) = 0$.

Solution. Let $X(s) = \mathcal{L}\{x(t)\}$ and $Y(s) = \mathcal{L}\{y(t)\}$. Transforming the system gives:

$$\begin{aligned}sX(s) - x(0) &= -Y(s) \implies sX(s) - 1 = -Y(s) \\ sY(s) - y(0) &= X(s) \implies sY(s) = X(s)\end{aligned}$$

This is a system of algebraic equations. Substituting the second equation into the first:

$$\begin{aligned}s(sY(s)) - 1 &= -Y(s) \implies s^2Y(s) + Y(s) = 1 \implies Y(s)(s^2 + 1) = 1 \\ Y(s) &= \frac{1}{s^2 + 1}\end{aligned}$$

Substituting this back to find $X(s)$:

$$X(s) = sY(s) = \frac{s}{s^2 + 1}$$

Finding the inverse transforms gives the solution pair:

$$\begin{aligned}y(t) &= \mathcal{L}^{-1}\left\{\frac{1}{s^2 + 1}\right\} = \sin(t) \\ x(t) &= \mathcal{L}^{-1}\left\{\frac{s}{s^2 + 1}\right\} = \cos(t)\end{aligned}$$

■

Example 4.1.4 (Volterra Integral Equation with Convolution). Solve the integral equation $y(t) = t + \int_0^t y(\tau) \sin(t - \tau) d\tau$.

Solution. We recognize the integral as the convolution $(y * \sin)(t)$. The equation is $y(t) = t + (y * \sin)(t)$. Taking the Laplace transform of the entire equation:

$$\mathcal{L}\{y(t)\} = \mathcal{L}\{t\} + \mathcal{L}\{(y * \sin)(t)\}$$

Using the Convolution Theorem, $\mathcal{L}\{(f * g)(t)\} = F(s)G(s)$:

$$Y(s) = \frac{1}{s^2} + Y(s) \cdot \mathcal{L}\{\sin(t)\} = \frac{1}{s^2} + Y(s) \frac{1}{s^2 + 1}$$

Solving for $Y(s)$:

$$\begin{aligned}Y(s) - Y(s) \frac{1}{s^2 + 1} &= \frac{1}{s^2} \implies Y(s) \left(1 - \frac{1}{s^2 + 1}\right) = \frac{1}{s^2} \\ Y(s) \left(\frac{s^2 + 1 - 1}{s^2 + 1}\right) &= \frac{1}{s^2} \implies Y(s) \frac{s^2}{s^2 + 1} = \frac{1}{s^2} \\ Y(s) &= \frac{s^2 + 1}{s^4} = \frac{1}{s^2} + \frac{1}{s^4}\end{aligned}$$

Taking the inverse transform, we recall $\mathcal{L}\{t^n\} = n!/s^{n+1}$:

$$y(t) = \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + \frac{1}{3!} \mathcal{L}^{-1}\left\{\frac{3!}{s^4}\right\} = t + \frac{1}{6}t^3$$

■

Example 4.1.5 (Integro-Differential Equation). Solve the integro-differential equation $y' + 2y + 5 \int_0^t y(\tau) d\tau = 1$ with the initial condition $y(0) = 0$.

Solution. Taking the Laplace transform of each term:

$$\begin{aligned}\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} + 5\mathcal{L}\left\{\int_0^t y(\tau) d\tau\right\} &= \mathcal{L}\{1\} \\ [sY(s) - y(0)] + 2Y(s) + 5\frac{Y(s)}{s} &= \frac{1}{s}\end{aligned}$$

Substituting $y(0) = 0$ and multiplying the entire equation by s to clear the fraction:

$$s^2Y(s) + 2sY(s) + 5Y(s) = 1$$

$$Y(s)(s^2 + 2s + 5) = 1 \implies Y(s) = \frac{1}{s^2 + 2s + 5}$$

To find the inverse transform, we complete the square in the denominator:

$$s^2 + 2s + 5 = (s^2 + 2s + 1) + 4 = (s + 1)^2 + 2^2$$

So, $Y(s) = \frac{1}{(s+1)^2+2^2}$. We adjust this to match the sine transform form $\frac{b}{(s-a)^2+b^2}$:

$$Y(s) = \frac{1}{2} \cdot \frac{2}{(s+1)^2 + 2^2}$$

The inverse transform is found using the s-Shifting Theorem with $a = -1$ and $b = 2$:

$$y(t) = \frac{1}{2}e^{-t}\sin(2t)$$

■

Example 4.1.6 (ODE with a Dirac Delta Forcing Function). Solve the initial value problem $y'' + 9y = \delta(t - 2)$, with $y(0) = 0$ and $y'(0) = 0$.

Solution. This models an oscillator at rest that receives a sharp "kick" or impulse at time $t = 2$. Transforming the equation:

$$\begin{aligned} [s^2Y(s) - sy(0) - y'(0)] + 9Y(s) &= \mathcal{L}\{\delta(t - 2)\} \\ s^2Y(s) + 9Y(s) &= e^{-2s} \end{aligned}$$

Solving for $Y(s)$:

$$Y(s)(s^2 + 9) = e^{-2s} \implies Y(s) = e^{-2s} \frac{1}{s^2 + 9}$$

We find the inverse transform using the Second Translation (t-Shifting) Theorem, which states that $\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t - a)H(t - a)$. Here, $a = 2$ and $F(s) = \frac{1}{s^2+9} = \frac{1}{3} \frac{3}{s^2+3^2}$. The inverse of $F(s)$ is $f(t) = \frac{1}{3}\sin(3t)$. Applying the theorem, the solution is:

$$y(t) = f(t - 2)H(t - 2) = \frac{1}{3}\sin(3(t - 2))H(t - 2)$$

This means the oscillator remains at rest until $t = 2$, after which it oscillates with an amplitude of $1/3$.

■

4.2 Applications to Boundary Value Problems

Example 4.2.1 (Heat Conduction with Initial Temperature). A semi-infinite rod ($x \geq 0$) has a uniform initial temperature of u_0 . At time $t = 0$, the end at $x = 0$ is instantaneously cooled to and maintained at a temperature of 0. Find the temperature $u(x, t)$ in the rod for $t > 0$.

Solution. The problem is described by the one-dimensional heat equation and its associated conditions:

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}, \quad x > 0, t > 0$$

- Initial Condition (IC): $u(x, 0) = u_0$
- Boundary Conditions (BCs): $u(0, t) = 0$, and $u(x, t)$ remains bounded as $x \rightarrow \infty$.

Let $U(x, s) = \mathcal{L}\{u(x, t)\}$. We transform the PDE with respect to time t :

$$sU(x, s) - u(x, 0) = k \frac{d^2U}{dx^2} \implies sU(x, s) - u_0 = k \frac{d^2U}{dx^2}$$

This rearranges into a non-homogeneous second-order ODE for $U(x, s)$:

$$\frac{d^2U}{dx^2} - \frac{s}{k}U(x, s) = -\frac{u_0}{k}$$

The general solution is the sum of the complementary function and a particular solution. The particular solution is constant, $U_p = \frac{u_0}{s}$. The complementary solution is $U_c = C_1e^{\sqrt{s/k}x} + C_2e^{-\sqrt{s/k}x}$. Thus, the general solution is $U(x, s) = C_1e^{\sqrt{s/k}x} + C_2e^{-\sqrt{s/k}x} + \frac{u_0}{s}$. We now apply the transformed boundary conditions: $\mathcal{L}\{u(0, t)\} = U(0, s) = 0$ and $\lim_{x \rightarrow \infty} U(x, s)$ must be bounded.

- For $U(x, s)$ to be bounded as $x \rightarrow \infty$, the coefficient of the growing exponential term must be zero, so $C_1 = 0$.
- At $x = 0$, we have $U(0, s) = C_2 + \frac{u_0}{s} = 0$, which implies $C_2 = -\frac{u_0}{s}$.

The solution in the s -domain is:

$$U(x, s) = \frac{u_0}{s} - \frac{u_0}{s} e^{-\frac{x}{\sqrt{k}}\sqrt{s}}$$

We find the inverse transform term-by-term. We know $\mathcal{L}^{-1}\{u_0/s\} = u_0$ and from standard tables, $\mathcal{L}^{-1}\left\{\frac{e^{-a\sqrt{s}}}{s}\right\} = \operatorname{erfc}\left(\frac{a}{2\sqrt{t}}\right)$, where erfc is the complementary error function. Here, $a = x/\sqrt{k}$. The inverse transform is:

$$u(x, t) = u_0 - u_0 \operatorname{erfc}\left(\frac{x}{2\sqrt{kt}}\right) = u_0 \left[1 - \operatorname{erfc}\left(\frac{x}{2\sqrt{kt}}\right)\right]$$

Since the error function $\operatorname{erf}(z) = 1 - \operatorname{erfc}(z)$, the solution is elegantly written as:

$$u(x, t) = u_0 \operatorname{erf}\left(\frac{x}{2\sqrt{kt}}\right)$$

■

Example 4.2.2 (Wave Propagation with a Step Input). A semi-infinite elastic string ($x \geq 0$), initially at rest and in its equilibrium position, has its end at $x = 0$ abruptly displaced by a constant amount u_0 and held there. Find the displacement $u(x, t)$.

Solution. The governing equation is the one-dimensional wave equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad x > 0, t > 0$$

- ICs: $u(x, 0) = 0$ and $u_t(x, 0) = 0$ (initially at rest)
- BCs: $u(0, t) = u_0$ for $t \geq 0$, and $u(x, t)$ is bounded as $x \rightarrow \infty$.

Let $U(x, s) = \mathcal{L}\{u(x, t)\}$. Transforming the PDE with respect to t :

$$s^2 U(x, s) - su(x, 0) - u_t(x, 0) = c^2 \frac{d^2 U}{dx^2}$$

Substituting the zero initial conditions simplifies this to a homogeneous ODE in x :

$$\frac{d^2 U}{dx^2} - \left(\frac{s}{c}\right)^2 U = 0$$

The general solution is $U(x, s) = C_1 e^{(s/c)x} + C_2 e^{-(s/c)x}$. The transformed boundary conditions are $U(0, s) = \mathcal{L}\{u_0\} = u_0/s$ and $\lim_{x \rightarrow \infty} U(x, s)$ is bounded.

- Boundedness requires $C_1 = 0$.
- At $x = 0$, $U(0, s) = C_2 = u_0/s$.

The solution in the s -domain is:

$$U(x, s) = \frac{u_0}{s} e^{-(x/c)s}$$

This form is a classic application of the Second Translation (t-Shifting) Theorem: $\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t-a)H(t-a)$. Here, $a = x/c$ and $F(s) = u_0/s$, so $f(t) = u_0$. The inverse transform is:

$$u(x, t) = u_0 H\left(t - \frac{x}{c}\right)$$

where H is the Heaviside step function. This solution represents a wavefront of height u_0 traveling to the right at speed c . The string at position x remains at rest until the wave reaches it at time $t = x/c$. ■

Example 4.2.3 (Voltage in a Lossless Transmission Line). A semi-infinite, lossless transmission line ($x \geq 0$) is initially uncharged. At $t = 0$, a constant voltage V_0 is applied at the sending end ($x = 0$). Find the voltage $v(x, t)$ along the line.

Solution. The voltage on a lossless line is governed by the wave equation, where the wave speed is $c = 1/\sqrt{LC}$ (L and C are inductance and capacitance per unit length).

$$\frac{\partial^2 v}{\partial t^2} = c^2 \frac{\partial^2 v}{\partial x^2}, \quad x > 0, t > 0$$

- ICs: $v(x, 0) = 0$ and $v_t(x, 0) = 0$ (initially uncharged and no current)

- BCs: $v(0, t) = V_0$ for $t \geq 0$, and $v(x, t)$ is bounded as $x \rightarrow \infty$.

This problem is mathematically identical to the previous example (Example 2). Following the exact same procedure, we transform the PDE and boundary conditions, solve the resulting ODE for the transformed voltage $V(x, s)$, and find:

$$V(x, s) = \frac{V_0}{s} e^{-(x/c)s}$$

Taking the inverse Laplace transform gives the voltage along the line:

$$v(x, t) = V_0 H\left(t - \frac{x}{c}\right)$$

This describes a step voltage of magnitude V_0 propagating down the transmission line at speed c . ■

Example 4.2.4 (Heat Conduction with a Ramp Temperature at the Boundary). A semi-infinite rod ($x \geq 0$) is initially at zero temperature. At $t = 0$, the temperature at the end $x = 0$ is increased linearly with time, such that $u(0, t) = At$ for some constant A . Find the temperature $u(x, t)$.

Solution. The problem setup is:

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}, \quad x > 0, t > 0$$

- IC: $u(x, 0) = 0$
- BCs: $u(0, t) = At$, and $u(x, t)$ is bounded as $x \rightarrow \infty$.

Let $U(x, s) = \mathcal{L}\{u(x, t)\}$. Transforming the PDE and using the zero initial condition gives the familiar ODE:

$$\frac{d^2 U}{dx^2} - \frac{s}{k} U(x, s) = 0$$

The general solution that remains bounded as $x \rightarrow \infty$ is $U(x, s) = C e^{-\sqrt{s/k}x}$. The transformed boundary condition at $x = 0$ is $U(0, s) = \mathcal{L}\{At\} = A/s^2$. Applying this condition gives $C = A/s^2$. The solution in the s -domain is:

$$U(x, s) = \frac{A}{s^2} e^{-\frac{x}{\sqrt{k}}\sqrt{s}}$$

Finding the inverse of this expression is more advanced and requires tables of transform pairs. A standard pair is:

$$\mathcal{L}^{-1} \left\{ \frac{e^{-a\sqrt{s}}}{s^2} \right\} = \left(t + \frac{a^2}{2} \right) \text{erfc} \left(\frac{a}{2\sqrt{t}} \right) - a \sqrt{\frac{t}{\pi}} e^{-a^2/4t}$$

Using $a = x/\sqrt{k}$, we obtain the temperature distribution:

$$u(x, t) = A \left[\left(t + \frac{x^2}{2k} \right) \text{erfc} \left(\frac{x}{2\sqrt{kt}} \right) - x \sqrt{\frac{t}{k\pi}} e^{-x^2/4kt} \right]$$

This demonstrates how the Laplace transform method can handle more complex boundary conditions, even if the final inversion step relies on known, non-trivial transform pairs. ■

4.3 Applications in Engineering and Physics

Example 4.3.1 (Electrical Circuits: RLC Series Circuit). An RLC series circuit has an inductor of $L = 1$ H, a resistor of $R = 20 \Omega$, a capacitor of $C = 0.01$ F, and a constant voltage source of $E = 12$ V. The circuit is switched on at $t = 0$. Find the current $i(t)$ in the circuit, assuming the initial current and initial charge on the capacitor are both zero.

Solution. The governing equation for an RLC circuit from Kirchhoff's Voltage Law is:

$$L \frac{di}{dt} + Ri + \frac{1}{C} \int_0^t i(\tau) d\tau = E(t)$$

Substituting the given values:

$$\frac{di}{dt} + 20i + 100 \int_0^t i(\tau) d\tau = 12$$

The initial condition is $i(0) = 0$. Let $I(s) = \mathcal{L}\{i(t)\}$. We take the Laplace transform of the equation:

$$[sI(s) - i(0)] + 20I(s) + 100 \frac{I(s)}{s} = \frac{12}{s}$$

Substituting $i(0) = 0$ and multiplying by s to clear the fractions:

$$s^2 I(s) + 20s I(s) + 100 I(s) = 12$$

Solving for $I(s)$:

$$I(s)(s^2 + 20s + 100) = 12 \implies I(s) = \frac{12}{(s + 10)^2}$$

To find the inverse transform, we use the s-Shifting Theorem. We know that $\mathcal{L}^{-1}\{1/s^2\} = t$.

$$i(t) = 12\mathcal{L}^{-1}\left\{\frac{1}{(s + 10)^2}\right\} = 12e^{-10t}\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\}$$

$$i(t) = 12te^{-10t} \text{ Amperes}$$

This represents a critically damped response where the current rises to a peak and then decays to zero. ■

Example 4.3.2 (Dynamics: Damped Forced Spring-Mass System). A mass of $m = 1$ kg is attached to a spring with constant $k = 5$ N/m. The system is damped with a damping coefficient of $b = 4$ Ns/m. The mass is initially at rest at its equilibrium position. At $t = 0$, it is subjected to an external force $F(t) = e^{-2t}$ N. Find the displacement $x(t)$ of the mass.

Solution. The equation of motion for a spring-mass-damper system is $mx'' + bx' + kx = F(t)$. Substituting the given values:

$$x'' + 4x' + 5x = e^{-2t}$$

The initial conditions are $x(0) = 0$ and $x'(0) = 0$. Let $X(s) = \mathcal{L}\{x(t)\}$. Transforming the equation:

$$[s^2 X(s) - sx(0) - x'(0)] + 4[sX(s) - x(0)] + 5X(s) = \mathcal{L}\{e^{-2t}\}$$

$$s^2 X(s) + 4sX(s) + 5X(s) = \frac{1}{s + 2}$$

Solving for $X(s)$:

$$X(s)(s^2 + 4s + 5) = \frac{1}{s + 2} \implies X(s) = \frac{1}{(s + 2)(s^2 + 4s + 5)}$$

We use partial fraction decomposition: $X(s) = \frac{A}{s+2} + \frac{Bs+C}{s^2+4s+5}$. Solving for the constants gives $A = 1$, $B = -1$, and $C = -2$.

$$X(s) = \frac{1}{s + 2} - \frac{s + 2}{s^2 + 4s + 5}$$

To invert the second term, we complete the square: $s^2 + 4s + 5 = (s + 2)^2 + 1^2$.

$$X(s) = \frac{1}{s + 2} - \frac{s + 2}{(s + 2)^2 + 1^2}$$

We now find the inverse transform of each term using the s-Shifting Theorem:

$$x(t) = \mathcal{L}^{-1}\left\{\frac{1}{s + 2}\right\} - \mathcal{L}^{-1}\left\{\frac{s + 2}{(s + 2)^2 + 1^2}\right\}$$

$$x(t) = e^{-2t} - e^{-2t} \cos(t) \text{ meters}$$
■

Example 4.3.3 (Beams: Cantilever Beam with a Point Load). A cantilever beam of length L and flexural rigidity EI is fixed at $x = 0$. It is subjected to a single concentrated downward load P at its free end, $x = L$. Find the deflection $y(x)$ of the beam.

Solution. The governing equation for beam deflection is $EI \frac{d^4 y}{dx^4} = w(x)$, where $w(x)$ is the distributed load. A point load can be represented using the Dirac delta function: $w(x) = P\delta(x - L)$. The equation becomes:

$$EI \frac{d^4 y}{dx^4} = P\delta(x - L)$$

The boundary conditions for a cantilever beam are $y(0) = 0$ (zero deflection at the fixed end) and $y'(0) = 0$ (zero slope at the fixed end). We use the Laplace transform with respect to the spatial variable x . Let $Y(s) = \mathcal{L}\{y(x)\}$. We also need to define the unknown shear and bending moment at the fixed end: $y''(0) = C_1$ and $y'''(0) = C_2$. Transforming the equation:

$$EI[s^4 Y(s) - s^3 y(0) - s^2 y'(0) - s y''(0) - y'''(0)] = P\mathcal{L}\{\delta(x - L)\}$$

$$EI[s^4 Y(s) - sC_1 - C_2] = Pe^{-Ls}$$

Solving for $Y(s)$:

$$Y(s) = \frac{C_1}{s^3} + \frac{C_2}{s^4} + \frac{P}{EI} \frac{e^{-Ls}}{s^4}$$

The inverse transform gives: $y(x) = C_1 \frac{x^2}{2} + C_2 \frac{x^3}{6} + \frac{P}{EI} \frac{(x-L)^3}{6} H(x-L)$. To find C_1 and C_2 , we use the boundary conditions at the free end ($x = L$): zero bending moment ($y''(L) = 0$) and zero shear force ($y'''(L) = 0$). $y''(x) = C_1 + C_2 x + \frac{P}{EI} (x-L) H(x-L) \Rightarrow y''(L) = C_1 + C_2 L = 0$. $y'''(x) = C_2 + \frac{P}{EI} H(x-L) \Rightarrow y'''(L) = C_2 + \frac{P}{EI} = 0$. From the second equation, $C_2 = -P/EI$. Substituting into the first gives $C_1 - PL/EI = 0 \Rightarrow C_1 = PL/EI$. Substituting the constants back into the solution for $y(x)$:

$$y(x) = \frac{PL}{EI} \frac{x^2}{2} - \frac{P}{EI} \frac{x^3}{6} = \frac{P}{6EI} (3Lx^2 - x^3) \quad \text{for } 0 \leq x < L$$

■

Example 4.3.4 (Heat Conduction: Semi-Infinite Rod with Heat Flux). A semi-infinite rod ($x \geq 0$) is initially at zero temperature. At $t = 0$, a constant heat flux q_0 is applied at the end $x = 0$. Find the temperature $u(x, t)$.

Solution. The problem is governed by the heat equation $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$. The boundary condition related to heat flux is given by Fourier's Law of Conduction, $q = -K \frac{\partial u}{\partial x}$, where K is thermal conductivity. So, at $x = 0$, we have $-K \frac{\partial u}{\partial x}(0, t) = q_0$. The conditions are:

- IC: $u(x, 0) = 0$
- BCs: $\frac{\partial u}{\partial x}(0, t) = -\frac{q_0}{K}$, and $u(x, t)$ is bounded as $x \rightarrow \infty$.

Let $U(x, s) = \mathcal{L}\{u(x, t)\}$. The transformed PDE is $\frac{d^2 U}{dx^2} - \frac{s}{k} U = 0$. The bounded solution is $U(x, s) = C e^{-\sqrt{s/k}x}$. The transformed flux boundary condition is $\frac{dU}{dx}(0, s) = \mathcal{L}\{-\frac{q_0}{K}\} = -\frac{q_0}{Ks}$. We differentiate $U(x, s)$ with respect to x : $\frac{dU}{dx} = -C \sqrt{s/k} e^{-\sqrt{s/k}x}$. At $x = 0$: $\frac{dU}{dx}(0, s) = -C \sqrt{s/k} = -\frac{q_0}{Ks}$. Solving for C : $C = \frac{q_0 \sqrt{k}}{K} \frac{1}{s \sqrt{s}} = \frac{q_0 \sqrt{k}}{K} s^{-3/2}$. So, $U(x, s) = \frac{q_0 \sqrt{k}}{K} \frac{e^{-\frac{x}{\sqrt{k}} \sqrt{s}}}{s^{3/2}}$. This is a known, non-trivial inverse transform pair: $\mathcal{L}^{-1}\left\{\frac{e^{-a\sqrt{s}}}{s^{3/2}}\right\} = 2\sqrt{\frac{t}{\pi}} e^{-a^2/4t} - a \operatorname{erfc}\left(\frac{a}{2\sqrt{t}}\right)$. With $a = x/\sqrt{k}$:

$$u(x, t) = \frac{q_0 \sqrt{k}}{K} \left[2\sqrt{\frac{t}{\pi}} e^{-x^2/4kt} - \frac{x}{\sqrt{k}} \operatorname{erfc}\left(\frac{x}{2\sqrt{kt}}\right) \right]$$

■

Example 4.3.5 (Wave Equation: Plucked String). A semi-infinite string ($x \geq 0$) is lifted to an initial displacement $u(x, 0) = A e^{-bx}$ and released from rest ($u_t(x, 0) = 0$). Find the subsequent motion $u(x, t)$.

Solution. The motion is governed by $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$.

- ICs: $u(x, 0) = A e^{-bx}$, $u_t(x, 0) = 0$.
- BCs: $u(0, t)$ is bounded (implied, no explicit condition needed here for the method).

Let $U(x, s) = \mathcal{L}\{u(x, t)\}$. Transforming the PDE gives:

$$s^2 U(x, s) - s u(x, 0) - u_t(x, 0) = c^2 \frac{d^2 U}{dx^2}$$

$$s^2 U - s A e^{-bx} = c^2 \frac{d^2 U}{dx^2} \Rightarrow \frac{d^2 U}{dx^2} - \left(\frac{s}{c^2}\right) U = -\frac{sA}{c^2} e^{-bx}$$

This is a non-homogeneous ODE for $U(x, s)$. The particular solution has the form $U_p = C e^{-bx}$. Substituting gives $C(b^2 - s^2/c^2) = -sA/c^2$, so $C = \frac{sA}{s^2 - b^2 c^2}$. The bounded complementary solution is $U_c = D e^{-(s/c)x}$. So, $U(x, s) = D e^{-(s/c)x} + \frac{sA}{s^2 - (bc)^2} e^{-bx}$. The constant D must be chosen so that the solution reflects the physical reality. A direct inversion of this form is complex. However, d'Alembert's solution for an infinite string is $u(x, t) = \frac{1}{2}[f(x-ct) + f(x+ct)]$. For a semi-infinite string fixed at $x = 0$, we use an odd extension of the initial shape. In this case with a free end, we use an even extension. The solution is:

$$u(x, t) = \frac{A}{2} [e^{-b(x-ct)} H(x-ct) + e^{-b(x+ct)}]$$

This represents two exponential shapes, one traveling right and one traveling left (which reflects off the "virtual" string for $x < 0$).

■

Example 4.3.6 (Dynamics: Rotational Motion with Damping). A flywheel with moment of inertia $J = 10 \text{ kg}\cdot\text{m}^2$ is subject to a viscous damping torque of $B = 20 \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$. It is initially at rest. At $t = 0$, a constant torque $\tau = 100 \text{ N}\cdot\text{m}$ is applied. Find the angular velocity $\omega(t)$.

Solution. The equation for rotational motion is $J\frac{d\omega}{dt} + B\omega = \tau(t)$. Substituting the values:

$$10\frac{d\omega}{dt} + 20\omega = 100 \implies \frac{d\omega}{dt} + 2\omega = 10$$

The initial condition is $\omega(0) = 0$. Let $\Omega(s) = \mathcal{L}\{\omega(t)\}$. Transforming the equation:

$$[s\Omega(s) - \omega(0)] + 2\Omega(s) = \frac{10}{s}$$

$$s\Omega(s) + 2\Omega(s) = \frac{10}{s} \implies \Omega(s)(s+2) = \frac{10}{s}$$

$$\Omega(s) = \frac{10}{s(s+2)}$$

Using partial fractions, $\frac{10}{s(s+2)} = \frac{5}{s} - \frac{5}{s+2}$.

$$\Omega(s) = \frac{5}{s} - \frac{5}{s+2}$$

The inverse transform gives the angular velocity:

$$\omega(t) = \mathcal{L}^{-1}\left\{\frac{5}{s}\right\} - \mathcal{L}^{-1}\left\{\frac{5}{s+2}\right\} = 5 - 5e^{-2t} \text{ rad/s}$$

The flywheel's speed exponentially approaches a terminal velocity of 5 rad/s. ■